This is one of my two preparation documents for my oral exam, focused on geometric/combinatorial group theory.

Contents

Preliminaries **1**
Group-Theoretic Constructions . . . 1
Group Homology/Cohomology . . . 3
Nilpotent Groups . . . 4
Finitely Generated Nilpotent Groups . . . 5

Groups and Graphs **7**
Graph Theory Review . . . 7
Cayley Graphs . . . 7
Actions of Groups on Trees . . . 9
Ping-Pong Lemma . . . 11
Applications . . . 11

Quasi-Isometry **14**
Quasi-Isometry of Metric Spaces . . . 14
Quasi-Isometry of Groups . . . 16
Quasi-Geodesics . . . 18
Milnor–Schwarz Lemma . . . 18
Applications of the Milnor–Schwarz Lemma . . . 20
Dynamic Criterion for Quasi-Isometry . . . 22
Asymptotic Cones . . . 25
Growth Rates and Types . . . 27

Amenability of Groups **29**
Means on Groups . . . 29
Examples and Inheritance Properties . . . 30
Approximation Properties . . . 31
Quasi-Isometry Invariance . . . 34
More Analytic Characterizations . . . 36

Soficity **39**
The Law of the Group von Neumann Algebra . . . 39
Metric Ultraproducts . . . 40
Approximate Homomorphisms . . . 42
Cayley Graphs . . . 44

Preliminaries

Group-Theoretic Constructions

There are two major ways we may construct new groups out of old: extensions/semidirect products and pushouts/amalgamated free products. We will provide a brief overview of both of these.

Definition: Let Q and N be groups. We say a group G is an extension of Q and N if there is an exact sequence

$$1 \to N \to G \to Q \to 1.$$

If the extension splits — i.e., there is a section $Q \to N$ — then $G \cong Q \ltimes_\varphi N$ is a semidirect product for some $\varphi\colon Q \to \mathrm{Aut}(N)$.

Example:

- If the homomorphism $\varphi\colon Q \to \mathrm{Aut}(N)$ is trivial, then $N \rtimes_\varphi Q \cong N \times Q$ is a direct product.
- The map $\mathbb{Z}/2\mathbb{Z} \to \mathrm{Aut}(\mathbb{Z}/n\mathbb{Z})$ given by inversion yields the dihedral group

$$D_n \cong \mathbb{Z}/2\mathbb{Z} \ltimes \mathbb{Z}/n\mathbb{Z}.$$

- If G is any group, the lamplighter group of G is the semidirect product $\mathbb{Z} \ltimes G^{\mathbb{Z}}$ with homomorphism $\varphi\colon \mathbb{Z} \to \mathrm{Aut}\left(G^{\mathbb{Z}}\right)$ given by

$$\varphi(z) = \left((g_n)_{n\in\mathbb{Z}} \mapsto (g_{n+z})_{n\in\mathbb{Z}}\right).$$

- the wreath product of G and H, denoted $G \wr H$, is the semidirect product $H \ltimes G^H$ with $\varphi\colon H \to \mathrm{Aut}\left(G^H\right)$ given by the shift action

$$\varphi(h) = \left((g_k)_{k\in H} \mapsto (g_{kh})_{k\in H}\right).$$

Definition: Let A be a group, $\alpha_1\colon A \to G_1$, and $\alpha_2\colon A \to G_2$ group homomorphisms. We say a group G is a *pushout* of G_1 and G_2 over A with respect to α_1 and α_2 if there are homomorphisms $\beta_1\colon G_1 \to G$ and $\beta_2\colon G_2 \to G$ satisfying $\beta_1 \circ \alpha_1 = \beta_2 \circ \alpha_2$ and the universal property for pushouts.

That is, for every group H and all group homomorphisms $\varphi_1\colon G_1 \to H$ and $\varphi_2\colon G_2 \to H$ satisfying $\varphi_1 \circ \alpha_1 = \varphi_2 \circ \alpha_2$, there is exactly one homomorphism $\varphi\colon G \to H$ satisfying $\varphi \circ \beta_i = \varphi_i$ for each $i = 1, 2$. We denote the pushout by $G_1 *_A G_2$.

- If $A = \{e\}$ is the trivial group, then the pushout $G_1 *_A G_2 = G_1 * G_2$ is called the *free product* of G_1 and G_2.
- If α_1 and α_2 are injective, then $G_1 *_A G_2$ is called an amalgamated free product of G_1 and G_2 over A.

Example:

- The free group F_n on n letters is the n-fold free product $\mathbb{Z} * \cdots * \mathbb{Z}$.
- The infinite dihedral group $D_\infty \cong \mathbb{Z}/2\mathbb{Z} \ltimes \mathbb{Z}$ is isomorphic to the free product $\mathbb{Z}/2\mathbb{Z} * \mathbb{Z}/2\mathbb{Z}$.
- The matrix group $\mathrm{SL}_2(\mathbb{Z})$ is isomorphic to the amalgamated free product $\mathbb{Z}/6\mathbb{Z} *_{\mathbb{Z}/2\mathbb{Z}} \mathbb{Z}/4\mathbb{Z}$.
- If X and Y are pointed, non-empty, path-connected topological spaces with nonempty and path-connected intersection, then the fundamental group of their union (which is a type of pushout) is given by the amalgamated free product of their fundamental groups over the fundamental group of the intersection.

Theorem: Pushout groups exist and are unique up to isomorphism.

Proof. Take presentations

$$G_i = \left\langle \left\{ x_g : g \in G_i \right\} | R_i \right\rangle.$$

Then, the presentation

$$G := \left\langle \left\{ x_g : g \in G_1 \right\} \cup \left\{ x_g : g \in G_2 \right\} | \left\{ x_{\alpha_1(a)} x_{\alpha_2(a)}^{-1} : a \in A \right\} \cup R_1 \cup R_2 \right\rangle.$$

The group homomorphisms $\beta_i\colon G_i \to G$ taking $g \mapsto x_g$, along with the relations $\left\{ x_{\alpha_1(a)} x_{\alpha_2(a)}^{-1} : a \in A \right\}$, give $\beta_1 \circ \alpha_1 = \beta_2 \circ \alpha_2$. The universal property then follows from the universal property of generators and relations. □

Definition: Let G be a group, and let $A, B \subseteq G$ be isomorphic subgroups with isomorphism $\theta\colon A \to B$.

The *HNN extension* of G with respect to θ is the group

$$G*_\theta := \left\langle \left\{x_g : g \in G\right\} \cup \{t\} \middle| \left\{t^{-1}x_a t x_{\theta(a)}^{-1} : a \in A\right\} \cup R \right\rangle,$$

where R is the relations for G. We say that t is the stable letter of the HNN extension.

The HNN extension is the universal (in a sense) way to force isomorphic subgroups to be conjugate to each other.

Group Homology/Cohomology

Definition: Let R be a commutative unital ring. The *group ring* is the ring $R[G]$ with

- underlying abelian group $\bigoplus_{g\in G} Rg$, where the g are formal symbols corresponding to the elements of G;
- multiplication given by the R-bilinear extension of the group multiplication $(g,h) \mapsto gh$.

If G is a non-abelian group and R is non-trivial, then the group ring $R[G]$ is not commutative.

Definition: Let G be a group.

- The *simplicial resolution* of G, denoted $C_*(G)$, is the chain complex of $\mathbb{Z}$-modules whose chains are defined by

  $$C_n := \bigoplus_{G^{n+1}} \mathbb{Z},$$

  with boundary maps given by $\partial_n \colon C_n(G) \to C_{n-1}(G)$ taking

  $$\partial_n\left(\sum_{g\in G^{n+1}} a_g(g_0,\dots,g_n)\right) = \sum_{g\in G^n} a_g \sum_{j=0}^{n} (-1)^j \left(g_0,\dots,g_{j-1},g_{j+1},\dots,g_n\right).$$

- If A is a left $\mathbb{Z}[G]$-module, then the simplicial chain complex of G with A-coefficients is given by $C_n(G;A) = A \otimes_{\mathbb{Z}[G]} C_*(G)$ with boundary operator $\partial_*^A = \mathrm{id}_A \otimes_{\mathbb{Z}[G]} \partial_*$. The elements of $\ker\left(\partial_n^A\right)$ are called the n-cycles.

  The n-th homology of G with A-coefficients is the $\mathbb{Z}$-module given by

  $$H_n(G;A) := \frac{\ker\left(\partial_n^A \colon C_n(G;A) \to C_{n-1}(G;A)\right)}{\mathrm{im}\left(\partial_{n+1}^A \colon C_{n+1}(G;A) \to C_n(G;A)\right)}.$$

- Let A be a left $\mathbb{Z}[G]$-module. The simplicial cochain complex of G with A-coefficients is defined by

  $$C^*(G;A) := \mathrm{Hom}_{\mathbb{Z}[G]}(C_*(G),A)$$

  with coboundary operator ∂_A^* given by the A-dual of ∂_*. Elements of $\ker\left(\delta_A^n\right)$ are called the n-cocycles. The nth cohomology of G with A-coefficients is the $\mathbb{Z}$-module

  $$H^n(G;A) := \frac{\ker\left(\partial_A^n \colon C^n(G;A) \to C^{n+1}(G;A)\right)}{\mathrm{im}\left(\partial_A^{n-1} \colon C^{n-1}(G;A) \to C^n(G;A)\right)}.$$

Nilpotent Groups

Lemma: Let G be a group, let $N \trianglelefteq G$ be a normal subgroup, and let $H, K, L \leq G$ be subgroups. If $[[H,K],L]$ and $[[K,L],H]$ are contained in N, then so is $[[L,H],K]$.

Proof. The subgroups $[[H,K],L]$, $[[K,L],H]$, and $[[L,H],K]$ are generated by the commutators $[h,k,\ell]$, $[k,\ell,h]$, and $[\ell,h,k]$ respectively, where $h \in H$, $k \in K$, and $\ell \in L$. The Hall–Witt identity says that

$$\begin{aligned} e &= b^{-1}\left[a,b^{-1},c\right]b \\ &= c^{-1}\left[b,c^{-1},a\right]c \\ &= a^{-1}\left[c,a^{-1}b\right]a, \end{aligned}$$

so all of these commutators are contained in N. □

Definition: Let G be a group. The *lower central series* of G is the sequence $(\gamma_k(G))_{k\geq 1}$ of subgroups defined recursively by

$$\begin{aligned} \gamma_1(G) &:= G \\ \gamma_{k+1}(G) &:= [\gamma_k(G), G]. \end{aligned}$$

We say a group G is *nilpotent* if its lower central series is finite. That is, if there is some integer $k \geq 1$ such that $\gamma_k(G) = e$. The smallest such k is called the *nilpotence class* (or nilpotency class) of G.

Inductively, we observe that $\gamma_k(G)$ is characteristic in G with $\gamma_{k+1}(G) \subseteq \gamma_k(G)$, meaning that the lower central series is a descending normal series.

Lemma: Let G be a group. Then,

$$\left[\gamma_i(G), \gamma_j(G)\right] \subseteq \gamma_{i+j}(G).$$

Proof. We prove by induction on i. By the definition of γ_{i+1}, we have that $\left[\gamma_1(G), \gamma_j(G)\right] = \left[G, \gamma_j(G)\right] = \gamma_{j+1}(G)$.

Now, we show that

$$\left[\gamma_{i+1}(G), \gamma_j(G)\right] \subseteq \gamma_{i+j+1}(G).$$

First, we claim that given $a, b \in \gamma_{i+1}(G)$ and $c \in \gamma_j(G)$, we have

$$\begin{aligned} [ab,c] &= [a,c][b,c] \bmod \gamma_{i+j+1}(G) \\ \left[a^{-1},c\right] &= [a,c]^{-1} \bmod \gamma_{i+j+1}(G). \end{aligned}$$

Since $a \in \gamma_{i+1}(G) \subseteq \gamma_i(G)$, the induction hypothesis has $[a,c] \in \left[\gamma_i(G), \gamma_j(G)\right] \subseteq \left[\gamma_{i+j}(G)\right]$. Consequently, $[[a,c],b] \in \left[\gamma_{i+j}(G), \gamma_j(G)\right] \subseteq \left[\gamma_{i+j}(G), G\right] = \gamma_{i+j+1}(G)$.

By standard commutator identities, we have

$$\begin{aligned} [ab,c] &= [a,c][a,c,b][b,c] \\ &= [a,c][b,c]\Big([b,c]^{-1}[a,c,b][b,c]\Big). \end{aligned}$$

Since $\gamma_{i+j+1}(G)$ is normal, it follows that $[b,c]^{-1}[a,c,b][b,c] \in \gamma_{i+j+1}(G)$, so this gives our first desired result. Similarly, by taking $b = a^{-1}$, we obtain the second desired result.

Therefore, we only need to show that $[a,x,b] \in \gamma_{i+j+1}(G)$ For all $a \in \gamma_i(G)$, $x \in G$, and $b \in \gamma_j(G)$.

From the Hall–Witt identity, we may write

$$[a,x,b] = x^{-1}\Big(\left(a^{-1}\left[b,a^{-1}x^{-1}\right]a\right)^{-1}\left(b^{-1}\left[x^{-1},b^{-1}a\right]b\right)^{-1}\Big)x.$$

while by the inductive hypothesis, $\left[b, a^{-1}, x^{-1}\right] \in \left[\gamma_{i+j}(G), G\right] = \gamma_{i+j+1}(G)$, and $\left[x^{-1}, b^{-1}, a\right] \in \left[\gamma_{j+1}(G), \gamma_i(G)\right] \subseteq \gamma_{i+j+1}(G)$, seeing as $\gamma_{i+j+1}(G)$ is a normal subgroup (hence the entire conjugacy class must land in it if any conjugate lands in it). □

We say a descending series $G = G_1 \geq G_2 \geq \cdots \geq G_k \geq G_{k+1} \geq \cdots$ is *central* if $\left[G_i, G_j\right] \subseteq G_{i+j}$ for all $i, j \geq 1$. This condition implies that each G_i is normal in G.

Lemma: Let G be a group. Suppose $(G_k)_{k\geq 1}$ is a descending central series for G. Then, $G_i \supseteq \gamma_i(G)$ for each $i \geq 1$.

Proof. We proceed by induction. First, $G_1 = G = \gamma_1(G)$.

Suppose the statement holds for i. The defining property of a central series and the inductive hypothesis give

$$\begin{aligned} G_{i+1} &\supseteq [G_i, G] \\ &\supseteq [\gamma_i(G), G] \\ &= \gamma_{i+1}(G). \end{aligned}$$

□

In particular, this means that a group G is nilpotent if and only if it admits any finite descending central series.

Proposition: Let G be a nilpotent group with nilpotence class c. Then, any subgroup and any quotient of G is nilpotent of nilpotence class at most c.

Proof. If $H \leq G$ is a subgroup, inductively we have $\gamma_i(H) \subseteq \gamma_i(G)$.

If $\varphi\colon G \to \overline{G}$ is a surjective homomorphism, then we have $\varphi([H, K]) = [\varphi(H), \varphi(K)]$ for all subgroups $H, K \leq G$. In particular, this means that inductively we have $\gamma_{i+1}(\varphi(G)) = \varphi(\gamma_{i+1}(G))$. □

Definition: Let G be a group. The *upper central series* of G is the ascending series $(Z_k(G))_{k\geq 0}$, where

$$\begin{aligned} Z_0(G) &= e \\ Z_{i+1}(G) &= \pi_i^{-1}(Z(G/Z_i(G))), \end{aligned}$$

where $\pi_i\colon G \to G/Z_i(G)$ is the projection homomorphism, and $Z(H)$ is the center of a group H.

In particular, we have $Z(G/Z_i(G)) = Z_{i+1}(G)/Z_i(G)$.

If G admits a finite upper central series, then the descending central series $(Z_{k-i+1})_{i=1}^{k+1}$ is central.

Lemma: Let G be a group. Suppose G admits a finite central series $G = G_1 \geq G_2 \geq \cdots \geq G_r = \{e_G\}$. Then, $Z_i(G) \supseteq G_{r-i}$ for all $i \geq 0$. In particular, this means $Z_{r-1}(G) = G$.

Proof. We induct on i. First, observe that $Z_0(G) = \{e_G\} = G_r$.

We will show that $Z_{i+1}(G) \supseteq G_{r-i-1}$. That is, we have to show that $\pi_i(G_{r-i-1}) \subseteq Z(G/Z_i(G))$. This is equivalent to showing that $[G_{r-i-1}, G] \subseteq Z_i(G)$. Yet, this follows from the fact that the sequence $(G_i)_{i\geq 0}$ is central, whence $[G_{i-r-1}, G] \subseteq G_{r-i}$, and $G_{r-i} \subseteq Z_i(G)$ by the induction hypothesis. □

In particular, observe that $\gamma_i(G) \subseteq Z_{c-i}(G)$ if G is nilpotent of nilpotence class c.

Finitely Generated Nilpotent Groups

We start by considering torsion-free finitely generated nilpotent groups. That is, finitely generated nilpotent groups where no nontrivial element has finite order.

Lemma: Let G be a torsion free (not necessarily finitely generated) nilpotent group with upper central series $(Z_k(G))_{k\geq 0}$. Then, all the factors $Z_{i+1}(G)/Z_i(G)$ are torsion-free abelian groups.

Proof. We use induction on nilpotence class. If $c = 1$, then $Z_1(G)/Z_0(G) = G$ is abelian, so it is a torsion-free abelian group by assumption.

Suppose it holds true for all nilpotent groups of nilpotence class $(c-1)$. Let G have nilpotence class c. Set $H = G/Z_1(G)$. Then, we have $Z_i(H) = Z_{i+1}(G)/Z_i(G)$ by induction and the definition of the upper central series. In particular, H has nilpotence class $(c-1)$.

By the third isomorphism theorem, we have $Z_{i+1}(G)/Z_i(G) \cong (Z_{i+1}(G)/Z_1(G))/(Z_i(G)/Z_1(G)) \cong Z_i(H)/Z_{i-1}(H)$, so it suffices to show that H is torsion-free to complete the proof.

Let $a \in G$. Suppose there exists an integer $m \geq 1$ such that $a^m \in Z_1(G)$, and suppose toward contradiction that $a \notin Z_1(G)$. Notice that $Z_1(G) = Z(G)$, so by assumption there is some $b \in G$ such that $[a,b] \neq e$. Let $1 \leq i \leq c-1$ be the maximal index for which there exists some $b \in \gamma_i(G)$ such that $[a,b] \neq e$, where $(\gamma_i(G))_{i=1}^{c+1}$ is the lower central series. By the various commutator identities, we have

$$\begin{aligned} e &= [a^m, b] \\ &= \left[aa^{m-1}, b\right] \\ &= a^{1-m}[a,b]a^{m-1}\left[a^{m-1}, b\right]. \end{aligned}$$

Since $[a,b] \in \gamma_{i+1}(G)$, we must have that a commutes with $[a,b]$, so $a^{1-m}[a,b]a^{m-1} = [a,b]$, and we have $e = [a^m, b] = [a,b]\left[a^{m-1}b\right]$. Iterating this process gives $e = [a,b]^m$, but since G is torsion free, this gives that $[a,b] = e$, which is a contradiction of the definition of b. Thus, $a \in Z_1(G)$, hence $H = G/Z_1(G)$ is torsion-free. □

We will take the following result as a black box.

Theorem (Malcev's Theorem)**:** Let G be a finitely generated torsion-free nilpotent group. Then, there exists $n \geq 1$ such that G embeds into $\mathrm{UT}(n, \mathbb{Z})$, the group of upper unitriangular matrices over $\mathbb{Z}$.

Theorem: Let G be a finitely generated nilpotent group. There exists a finite descending normal series

$$G \geq N_1 \geq \cdots \geq N_s \geq N_{s+1} = \{e\}$$

such that N_k/N_{k+1} is cyclic and $[N_k, G] \subseteq N_{k+1}$ for all $k = 1, \ldots, s$.

Proof. Let $(\gamma_i)_{i \in I}$ be the lower central series of G. Since γ_i/γ_{i+1} is a finitely generated abelian group, there are normal subgroups

$$\gamma_i \geq N_{i,1} \geq N_{i,2} \geq \cdots \geq N_{i,t_i} = \gamma_{i+1}$$

with $N_{i,j}/N_{i,j+1}$ cyclic for each j. Additionally, we have

$$\begin{aligned} \left[N_{i,j}, G\right] &\subseteq [\gamma_i, \gamma_1] \\ &= \gamma_{i+1} \\ &= N_{i,t_i} \\ &\subseteq N_{i,j+1}. \end{aligned}$$

Renaming the normal subgroups yields the desired result. □

The next focus will be on torsion in finitely generated nilpotent groups.

Lemma: Every subgroup of a finitely generated nilpotent group is finitely generated.

Proof. Let G be a finitely generated nilpotent group with

$$G = N_1 \geq N_2 \geq \cdots \geq N_s \geq N_{s+1} = \{e\}$$

a normal series with N_i/N_{i+1} cyclic for each $i = 1, \ldots, s$. We will show that each subgroup $H \leq G$ is generated by at most s elements.

If $s = 1$, then G is cyclic and so the statement necessarily holds. Assume the statement holds for all groups with normal series with cyclic quotients of length $(s-1)$. Since G/N_2 is cyclic, so is the subgroup HN_2/N_2. By the second isomorphism theorem, this is isomorphic to $H/(H \cap N_2)$. Since $H \cap N_2$ is a subgroup of N_2,

the induction hypothesis gives that H is generated by at most $(s-1)$ elements, whence H is generated by at most s elements. □

Groups and Graphs

One of the first connections between groups and geometry is the notion of a Cayley graph, and the way that the action of a group on a graph is connected to the structure of the group.

Graph Theory Review

Definition:

- A graph is a pair $X = (V, E)$, where $E \subseteq V^{[2]} := \{e : e \subseteq V, |e| = 2\}$. Elements of E are denoted by $\{v_1, v_2\}$ if the graph is undirected. If the graph is directed, then elements of E are denoted by tuples (v_1, v_2).
- If (V, E) is a graph, then we say vertices $v, v' \in V$ are adjacent if $\{v, v'\} \in E$. The number of neighbors is the degree of the vertex.
- If $X = (V, E)$ and $X' = (V', E')$ are graphs, then we say X and X' are isomorphic if there is a graph isomorphism between X and X' — that is, a bijection $f : V \to V'$ such that for any $v, w \in V$, we have $\{v, w\} \in E$ if and only if $\{f(v), f(w)\} \in E'$.

Definition: Let $X = (V, E)$ be a graph.

- A path in X is a sequence $[v_0, \dots, v_n]$ of distinct vertices such that $\left\{v_j, v_{j+1}\right\} \in E$.
- A graph is called connected if every pair of its vertices can be connected by a path in X.
- A cycle of length n is a path $[v_0, \dots, v_n]$ such that $\{v_n, v_0\} \in E$.
- A tree is a connected graph without cycles. Note that this is equivalent to a graph where every pair of vertices has exactly one path connecting them.
- If X is a connected graph, a *spanning tree* for X is a tree that contains all the vertices of X.

In algebraic topology, we use spanning trees as a way to compute fundamental groups. This is done by using the fact that spanning trees are CW subcomplexes (viewing the graph as a CW complex).

Cayley Graphs

Definition: Let G be a group, and let $S \subseteq G$ be a generating set for G without the identity. The Cayley graph of G with respect to S is the graph $\mathrm{Cay}(G, S)$, with

- vertices given by elements of G;
- edges given by

$$\left\{\{g, gs\} : g \in G, s \in S \cup S^{-1}\right\}.$$

Example:

- The Cayley graph of $\mathbb{Z}$ with respect to $\{1\}$ appears as an infinite line, while the Cayley graph of $\mathbb{Z}$ with respect to the generating set $\{2, 3\}$ appears as a particular 4-regular graph. See [Löh17, Figure 3.3].
- The Cayley graph of $\mathbb{Z}^2 \subseteq \mathbb{R}^2$ with respect to the generating set $\{(1, 0), (0, 1)\}$ appears as the integer lattice.
- The Cayley graph of $\mathbb{Z}/6\mathbb{Z}$ with respect to $\{[1]\}$ is a cycle graph.
- The Cayley graph of the symmetric group S_3 with respect to the generating set $\{\sigma, \tau\}$, where we have

$$S_3 := \left\langle \sigma, \tau | \sigma^3, \tau^2, \tau\sigma\tau\sigma \right\rangle.$$

- The Cayley graph of any group with respect to the generating set given by the group itself is a complete graph.

One of the seemingly simple questions one might ask is when exactly finitely generated groups G and H with finite generating sets $S \subseteq G$ and $T \subseteq H$ are such that $\mathrm{Cay}(G,S) \cong \mathrm{Cay}(H,T)$ are isomorphic as graphs. This is essentially a question about Cayley graphs whose automorphisms are affine (i.e., given by a group isomorphism and translation by a fixed group element).

- Cayley graphs of finitely generated abelian groups satisfy the rigidity question; the automorphisms are affine on the free part, and Cayley graphs remember the rank of the group and the size of the torsion subgroup.
- Automorphisms of Cayley graphs of finitely generated torsion-free nilpotent groups are affine.
- Finitely generated free groups admit isomorphic Cayley graphs if and only if they have the same rank.

Given a presentation of a group, the presentation complex is a two-dimensional CW complex given by taking one zero-cell, adding circles for every generator, and attaching disks for every relation (attached by the word for the relation).

The presentation complex associated to $\left\langle x,y|xyx^{-1}y^{-1}\right\rangle$ is the 2-torus. The presentation complex associated to $\left\langle x|x^2\right\rangle$ is $\mathbb{RP}^2$.

In general, any group admits a classifying space (known as the Eilenberg–MacLane space) whose fundamental group is the underlying group and whose higher homotopy groups are trivial. These spaces are unique up to homotopy equivalence. For example, the torus is a classifying space for $\mathbb{Z}^2$ and $\mathbb{RP}^\infty$ is a classifying space for $\mathbb{Z}/2\mathbb{Z}$. The 1-skeleton of the universal cover of the presentation complex of the presentation $\langle X|R\rangle$ is basically the Cayley graph of the underlying group (save for generators of order 2).

Theorem: Let F be a free group freely generated by $S \subseteq F$. Then, $\mathrm{Cay}(F,S)$ is a tree.

Proof. Every element of F is a unique word in elements of S, so there is a unique path between the identity and any word, so there is a unique path between any two elements, so $\mathrm{Cay}(F,S)$ is a tree. □

There is a converse, but it requires a caveat. In particular, we see that $\mathrm{Cay}(\mathbb{Z}/2\mathbb{Z},[1])$ is a one-edge two-vertex graph (which is a tree).

Theorem: Let G be a group, $S \subseteq G$ a generating set satisfying $st \neq e$ for all $s,t \in S$. If $\mathrm{Cay}(G,S)$ is a tree, then $G \cong F(S)$ is freely generated by S.

Proof. Let S and G be such that $\mathrm{Cay}(G,S)$ is a tree. We will show that G is isomorphic to $F(S)$ by an isomorphism that is the identity on S.

First, we observe that, by the universal property applied to the inclusion $S \subseteq G$, there is a group homomorphism $\varphi\colon F(S) \to G$ that is the identity on S. Since S generates G, it follows that φ is surjective.

Assume toward contradiction that φ is not injective. Let $s_1 \cdots s_n \in F(S) \setminus \{\varepsilon\}$ be a word of minimal length mapped to $e \in G$ by φ.

First, since $\varphi|_S = \mathrm{id}_S$, we must have $n > 1$.

Next, if $n = 2$, then we would have

$$\begin{aligned} e &= \varphi(s_1 s_2) \\ &= \varphi(s_1)\varphi(s_2) \\ &= s_1 s_2, \end{aligned}$$

contradicting the fact that $s_1 \cdots s_n$ is reduced with $st \neq e$ in G.

Next, suppose $n \geq 3$. Inductively, we may consider the sequence of elements $g_0, \ldots, g_{n-1}$ in G given by $g_{j+1} = g_j s_{j+1}$ for all $0 \leq j \leq n-2$. By minimality of the word $s_1 \cdots s_n$, the elements $g_0, \ldots, g_{n-1}$ are distinct, and

since $g_n = s_1 \cdots s_n = e$, it follows that $[g_0, \ldots, g_{n-1}]$ is a cycle in $\mathrm{Cay}(G, S)$. Yet, this contradicts the fact that $\mathrm{Cay}(G, S)$ is a tree. Thus, φ is injective. □

It turns out that, generally speaking, we are able to reconstruct a Cayley graph from any action of a group on a graph.

Proposition: Let G be a group, and let G act on a connected graph $X = (V, E)$ by graph automorphisms. If this action is free and transitive on the set V of vertices of X, then for a distinguished vertex x, the set

$$S \coloneqq \{s \in G : \{x, s.x\} \in E\}$$

generates G, and $\mathrm{Cay}(G, S) \cong X$ are isomorphic as graphs.

Proof. Since the action of G on the vertices is free and transitive, we may identify V with G by the identification $g \mapsto g.x$.

We start by showing that S generates G. Let $g \in G$. Since the graph X is connected, there is a path $[x_0, x_1, \ldots, x_n]$, where $x_0 = x$ and $x_n = g.x$, that joins x and $g.x$. We will translate this path into steps in the group G as follows.

For each $0 \leq j \leq (n-1)$, define

$$g_j = \varphi^{-1}\left(x_j\right)$$

in G. Then, we take $s_j = g_j^{-1} g_{j+1}$. We will show that $s_j \in S$. Since $[x_0, \ldots, x_n]$ is a path in X, we have $\left\{x_j, x_{j+1}\right\}$ is an edge of X. Since G acts by graph automorphisms, we have that $\left\{g_j^{-1}.x_j, g_j^{-1}.x_{j+1}\right\}$ is also an edge of X, but this is equal to $\left\{x, \left(g_j^{-1} g_{j+1}\right).x\right\}$. In particular, this means that

$$\begin{aligned} g &= g_n \\ &= g_0 g_0^{-1} g_1 g_1^{-1} \cdots g_{n-1} g_{n-1}^{-1} g_n \\ &= s_0 s_1 \cdots s_{n-1}, \end{aligned}$$

so S is a generating set for G.

We only need to show now that $\mathrm{Cay}(G, S)$ is isomorphic to the graph X. We will show that φ induces a graph isomorphism. If $g, h \in G$, then we observe that $\{\varphi(g), \varphi(h)\} = \{g.x, h.x\}$. Since G acts by graph automorphisms on X, the set is an edge of X if and only if $\left\{x, \left(g^{-1}h\right).x\right\}$ is an edge of X, which holds if and only if $g^{-1}h \in S$, which is equivalent to $\{g, h\}$ being an edge of $\mathrm{Cay}(G, S)$. Thus, $\mathrm{Cay}(G, S)$ is isomorphic to X. □

Actions of Groups on Trees

Similar to how a group is a free group with generating set S if and only if $\mathrm{Cay}(G, S)$ is a tree, there is a converse direction. That is, a group is free if and only if it admits a free action on a non-empty tree.

One direction emerges immediately from the fact that left-translation is a free action of G on itself that acts on its Cayley graph.

The reverse direction is a little more difficult. If $G \curvearrowright T$ is an action, we will have to find a tree that is a Cayley graph for G.

Definition: Let G be a group that acts on a connected graph X by graph automorphisms. A *spanning tree* of this action is a subgraph of X that is both a tree and contains exactly one vertex of every orbit of the action of G on the vertices of X.

Theorem: Every action of a group on a connected graph by automorphisms has a spanning tree.

Proof. Let G be a group acting on a connected graph X. We may assume that X is nonempty.

Consider the set T_G of all subtrees of X that contain at most one vertex of every orbit. This is a nonempty set since X has at least one vertex. The set T_G is partially ordered by the subgraph relation, and every to-

tally ordered chain of T_G has an upper bound in T_G (namely, the union of the trees in this chain). Thus, by Zorn's Lemma, there is a maximal element T in T_G that is necessarily nonempty since X is nonempty.

Suppose toward contradiction that T is not a spanning tree for the action of G. Then, there is a vertex v such that none of the vertices of the orbit $G.v$ is a vertex of T. We will show that there is some such v that is adjacent to a vertex of T.

Since X is connected, there is a path p connecting some vertex u of T with v, which we will label as

$$p = [u, \dots, v', \dots, v],$$

where v' is the first vertex of p that is not in T.

(i) If none of the vertices of the orbit $G.v'$ are contained in T, then v' is the vertex with our desired property.

(ii) If there is some $g \in G$ such that $g.v'$ is a vertex of T, write a subpath $p' = [v', \dots, v]$ of p, and consider the path

$$g.p' = [g.v', \dots, g.v].$$

Notice that $g.v$ is not contained in T since none of the elements of the orbit of v are contained in T by assumption. Since the length of p' is smaller than the length of p, we may iterate this process to eventually obtain our desired vertex.

Having obtained our vertex v with none of the vertices of the orbit $G.v$ in T and some neighbor u of v is contained in T, we may add the edge $\{u, v\}$ to T to produce a tree in T_G that contains T as a proper subgraph. This yields the contradiction, as we had assumed T was maximal. Thus, T is a spanning tree for the action of G on X. □

In order to show the reverse direction, we will have to determine, given an action $G \curvearrowright T$, a set S such that $\mathrm{Cay}(G, S)$ is a tree.

Theorem: If a group G acts freely on a tree T, then G is a free group.

Proof. Let T' be a spanning tree for the action of T. Say two translations $g.T'$ and $h.T'$ are adjacent if $g.T' \neq h.T'$ and there exist vertices $v \in g.T'$ and $w \in h.T'$ such that $\{v, w\}$ is an edge of T. Define the graph X by saying $V(X)$ consists of the set $\{g.T' : g \in G\}$ and $E(X)$ consists of all the adjacent trees $\{g.T', h.T'\}$.

We will start by showing that if $g \neq h$, then $g.T' \neq h.T'$. We may reduce this question to showing that for any $g \neq e$, we have that $g.T' \neq T'$. Yet, this follows directly from our assumptions; there is exactly one representative from each orbit in the spanning tree, and we cannot have $g.v = v$ for any $v \in T'$ unless $g = e$ by the assumption of the freeness of the action G.

For any adjacent $g.T'$ and $h.T'$, we claim that there is a unique connecting edge. We observe that $g.T'$ and $h.T'$ are adjacent if and only if $h^{-1}g.T'$ is adjacent to T' — that is, there exists $u \in T'$ and $v \in h^{-1}g.T'$ such that $\{u, v\}$ is an edge. Yet, since u is a unique representative of the orbit of v, it follows that there is only one such edge. (In Löh's book, such an edge is dubbed an "essential edge" on page 89.)

We observe that, since each copy of T' is a tree (hence connected), and T itself is a tree, it follows that there can only be one unique path between any two vertices in T' as else that implies the existence of two non-unique paths in the original graph T.

Finally, since $V(X) \curvearrowleft G$ freely by taking $(h.T').g = (hg).T'$, we must have that $\mathrm{Cay}(G, S) \cong X$. Since X is a tree, it follows that G is a free group. □

The following is a more topological proof that relies heavily on covering space theory.

Proof via Covering Spaces. Let $G \curvearrowright T$ freely by graph automorphisms. We may view T as a one-dimensional CW complex; since T is a tree, this one-dimensional CW complex is simply connected, so the covering map $T \to T/G$ has it such that T is the universal cover for the CW complex T/G. In particular, we have that the deck transformations of T are exactly G, and $\pi_1(T/G) \cong G$.

Since any one-dimensional CW complex is homotopy-equivalent to a bouquet of circles, the Seifert–Van Kampen theorem gives that $\pi_1(T/G) \cong F(S)$ for some generating set S, so that G is free. □

We now focus on some applications.

Theorem (Nielsen–Schreier Theorem)**:** Let G be a free group, and let $H \leq G$ be a subgroup. Then, H is a free group.

Proof. If $G \curvearrowright T$ acts freely, then so too does H. □

Ping-Pong Lemma

In order to determine the existence of free subgroups of some group G, we use the much celebrated ping-pong lemma.

Theorem (Ping-Pong Lemma)**:** Let G_1 and G_2 be subgroups of a group G with $|G_1| \geq 3$ and $|G_2| \geq 2$. Suppose G is generated by the union $G_1 \cup G_2$. If there is a G-action on a set X such that there are nonempty subsets $X_1, X_2 \subseteq X$ with X_2 not contained in X_1 and such that for all $g \in G_i \setminus \{e\}$, we have $g.X_j \subseteq X_i$ when $i = 1, 2$ and $j = 2, 1$, then $G \cong G_1 * G_2$.

Proof. Suppose w is a nontrivial reduced word in G_1 and G_2; by contracting if necessary, we start by assuming that w is of the form $w = a_1 b_1 \cdots b_{k-1} a_k$ with $a_i \in G_1 \setminus \{e\}$ and $b_i \in G_2 \setminus \{e\}$. Our goal is to show that w is not the identity in g. Observe that we have

$$\begin{aligned} w(X_2) &= a_1 b_1 \cdots b_{k-1} a_k (X_2) \\ &\subseteq a_1 b_1 \cdots b_{k-1} (X_1) \\ &\vdots \\ &\subseteq a_1 b_1 (X_1) \\ &\subseteq a_1 (X_2) \\ &\subseteq X_1. \end{aligned}$$

Since $X_2 \not\subseteq X_1$, it follows that $w \neq e_G$. The other cases for the representation of the reduced word w follow from conjugating with an element of G_1. □

Applications

The first application we will consider is the matrix group $\mathrm{SL}_2(\mathbb{Z})$. The matrices

$$a = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$$
$$b = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$$

generate a subgroup of $\mathrm{SL}_2(\mathbb{Z})$ that is isomorphic to the free group F_2. Considering the subsets of $\mathbb{R}^2$ given by

$$A = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} : |x| > |y| \right\}$$
$$B = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} : |x| < |y| \right\}.$$

Then, we have

$$a^n \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + 2ny \\ y \end{pmatrix},$$

whose action on B satisfies

$$|x + 2ny| \geq |2ny| - |x|$$

$$\begin{aligned}
&\geq 2|y| - |x| \\
&> 2|y| - y \\
&= |y|,
\end{aligned}$$

so that $a^n.B \subseteq A$, and similarly $b^m.A \subseteq B$ for all $m, n \neq 0$. Thus, the subgroup $\langle a, b\rangle \subseteq \mathrm{SL}_2(\mathbb{Z})$ is freely generated by the ping-pong lemma.

Proposition: The subgroup $F = \langle a, b\rangle \subseteq \mathrm{SL}_2(\mathbb{Z})$ is finite index.

Specifically, considering subgroups

$$\begin{aligned}
G &= \left\{ \begin{pmatrix} 2m+1 & 2r \\ 2s & 2n+1 \end{pmatrix} : m, n, r, s \in \mathbb{Z}, \det \begin{pmatrix} 2m+1 & 2r \\ 2s & 2n+1 \end{pmatrix} = 1 \right\} \\
F' &= \left\{ \begin{pmatrix} 4m+1 & 2r \\ 2s & 4n+1 \end{pmatrix} : m, n, r, s \in \mathbb{Z}, \det \begin{pmatrix} 4m+1 & 2r \\ 2s & 4n+1 \end{pmatrix} = 1 \right\},
\end{aligned}$$

we have $[G : F'] = 2$, $[\mathrm{SL}_2(\mathbb{Z}) : G] = 6$, and $F = F'$.

Proof. First, we show that $[G : F'] = 2$. Taking modulo 4, we have either both m and n are odd or both m and n are even. If m and n are even, then an element $x \in G$ is contained in F', and if m and n are odd, then $-x \in F'$. Therefore, $\{gF' : g \in G\} = F' \sqcup (-1)F'$. Therefore, $[G : F'] = 2$.

Next, we show that $[\mathrm{SL}_2(\mathbb{Z}) : G] = 6$. Observe that G is the kernel of the reduction modulo 2 homomorphism $\mathrm{SL}_2(\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{Z}/2\mathbb{Z})$. Since $|\mathrm{SL}_2(\mathbb{Z}/2\mathbb{Z}) = 6|$, it follows that $[\mathrm{SL}_2(\mathbb{Z}) : G] = 6$.

Finally, we observe that $F \subseteq F'$ since any word in the matrices a and b is of the given form. Writing

$$\begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{pmatrix} \in F',$$

we use induction on $\min(|x_{11}|, |x_{12}|)$ to show that $x \in F$.

- If $x - 12 = 0$, then we must have $x_{11} = x_{22} \in \langle b\rangle \subseteq F$.
- If $|x_{12}| \geq |x_{11}|$, then we will find $k \in \mathbb{Z}$ and $0 \leq R \leq 2|x_{11}| - 1$, with

  $$x_{12} + |x_{11}| = -2kx_{11} + R.$$

  Then, the matrix

  $$\begin{aligned}
  x' &= xa^k \\
  &= \begin{pmatrix} x_{11} & x_{12} + 2kx_{11} \\ x_{21} & x_{22} + 2kx_{21} \end{pmatrix}.
  \end{aligned}$$

  We have that $x' \in xF$, and

  $$\begin{aligned}
  |x'_{12}| &= |x_{12} + 2kx_{11}| \\
  &= |-|x_{11}| - 2kx_{11} + R + 2kx_{11}| \\
  &\leq |x_{11}| \\
  &= |x'_{11}|.
  \end{aligned}$$

  By parity, we have that $|x'_{12}| \neq |x'_{11}|$, so that $|x'_{12}| < |x'_{11}|$, meaning that $\min\left(|x'_{11}|, |x'_{12}|\right) < \min(|x_{11}|, |x_{12}|)$, whence $x' \in F$, meaning $x \in F$.
- Similarly, if $|x_{12}| < |x_{11}|$, then we will find $k \in \mathbb{Z}$ with $0 \leq R \leq |2x_{12}| - 1$ such that $x_{11} + |x_{12}| = -2kx_{12} + R$.

  Considering the matrix $x' = xb^k \in xF$ yields the same result with $\min\left(|x'_{11}|, |x'_{12}|\right) < \min(|x_{11}|, |x_{12}|)$ by a similar process as the previous item.

□

We note that it is possible to extend this method to show that, if we let

$$G_1 = \left\langle \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix} \right\rangle$$
$$G_2 = \left\langle \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \right\rangle$$
$$A = G_1 \cap G_2,$$

then $\mathrm{SL}_2(\mathbb{Z}) \cong G_1 *_A G_2$. Since $G \cong \mathbb{Z}/6\mathbb{Z}$, $G_2 \cong \mathbb{Z}/4\mathbb{Z}$, and $A \cong \mathbb{Z}/2\mathbb{Z}$, it follows that $\mathrm{SL}_2(\mathbb{Z}) \cong \mathbb{Z}/6\mathbb{Z} *_{\mathbb{Z}/2\mathbb{Z}} \mathbb{Z}/4\mathbb{Z}$.

The next application is about regular graphs of large girth. Recall that if X is a graph, then the girth $g(X)$ is the length of the shortest cycle in X. A tree (and/or a forest) has infinite girth.

Finding graphs of large girth, especially those with minimal vertex count, is a very important goal, and the solution can be found through the group-theoretic constructions we are interested in. We will focus on the specific case of 4-regular graphs.

Theorem: Let $N \geq 5$. Then, there exists a graph X_N with the following properties:

- X_N is 4-regular;
- X_N has at most N^3 vertices;
- the graph X_N satisfies

$$g(X_N) \geq 2\frac{\ln(N/2)}{\ln\left(1+\sqrt{2}\right)} - 1.$$

Proof. Let

$$a = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$$
$$b = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$$

be the matrices in $\mathrm{SL}_2(\mathbb{Z})$ that generate a free subgroup of rank 2. Then, we consider the homomorphism $\varphi_N \colon \mathrm{SL}_2(\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$ given by reduction modulo N. Let $a_N = \varphi_N(a)$ and $b_N = \varphi_N(b)$.

Let $G_N = \langle a_N, b_N \rangle$ and $X_N = \mathrm{Cay}(G, \{a_N, b_N\})$. Then, X_N is 4-regular by construction, and X_N has at most $|\mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})| \leq N^3$ vertices.

Thus, our task is to show the lower girth bound. Let v, w be reduced words in the free group $F(a,b)$ with $w \neq v$ and $\varphi_N(w) = \varphi_N(v)$. We may assume that the lengths m and n of w and v respectively satisfy

$$g(X_N) = m + n$$
$$\max(m,n) \leq \frac{m+n+1}{2}.$$

Let $c = w - v \in \mathrm{Mat}_2(\mathbb{Z})$. Since $w \neq v$, it follows that $c \neq 0$, but $c_N = \varphi_N(c)$, given by reduction modulo N, we have that $c_N = 0$. Therefore, we have $c' \in \mathrm{Mat}_2(\mathbb{Z})$ such that $c = Nc'$. Therefore, we have estimates of the operator norm given by

$$\begin{aligned} \|w\| + \|v\| &\geq \|c\| \\ &= N\left\|c'\right\| \\ &\geq N \max\left(\left|c'_{11}\right|, \left|c'_{12}\right|, \left|c'_{21}\right|, \left|c'_{22}\right|\right) \\ &\geq N. \end{aligned}$$

Yet, we already have that $\|a\| \leq 1 + \sqrt{2} =: d$, and $\|b\| \leq d$ as well. Therefore, we have

$$\begin{aligned} N &\leq \|c\| \\ &\leq d^m + d^n \\ &\leq 2d^{\frac{m+n+1}{2}} \\ &= 2d^{\frac{g(X_N)+1}{2}}, \end{aligned}$$

giving our desired result. □

Quasi-Isometry

Quasi-Isometry of Metric Spaces

Definition: Consider two metric spaces (X, d) and (Y, ρ), and a map $f : X \to Y$.

- We say f is an isometric embedding if for all $x, x' \in X$, $\rho(f(x), f(x')) = d(x, x')$.
- We say f is an isometry (or isometric isomorphism) if f is an isometric embedding, and there exists an isometric embedding $g : Y \to X$ satisfying $f \circ g = \mathrm{id}_Y$ and $g \circ f = \mathrm{id}_X$.
- We say f is a bilipschitz embedding if there is some constant $C > 0$ such that for all $x, x' \in X$,

$$\frac{1}{C} d(x, x') \leq \rho(f(x), f(x')) \leq C d(x, x').$$

- We say f is a bilipschitz equivalence if it is a bilipschitz embedding and there exists a bilipschitz embedding $g : Y \to X$ satisfying $f \circ g = \mathrm{id}_Y$, $g \circ f = \mathrm{id}_X$. The spaces X and Y are then called bilipschitz equivalent.
- The map f is called a (C, K)-quasi-isometric embedding if for all $x, x' \in X$, we have

$$\frac{1}{C} d(x, x') - K \leq \rho(f(x), f(x')) \leq C d(x, x') + K.$$

- The map $f' : X \to Y$ is said to have finite distance from f if there is some $K > 0$ such that for all $x \in X$, we have $\rho(f(x), f'(x)) \leq K$.
- The map f is called a quasi-isometry if there is a quasi-inverse quasi-isometric embedding $g : Y \to X$. That is, g is a quasi-isometric embedding with $g \circ f$ having finite distance from id_X and $f \circ G$ having finite distance from id_Y. We say X and Y are quasi-isometric, and write $X \sim_{\mathrm{QI}} Y$.

Proposition: A map $f : (X, d) \to (Y, \rho)$ is a quasi-isometry if and only if it is a quasi-isometric embedding with quasi-dense image. A map is said to have quasi-dense image if there is some constant C such that for all $y \in Y$, there is some $x \in X$ such that

$$\rho(f(x), y) \leq C.$$

Proof. If f is a quasi-isometry, then there is a quasi-inverse quasi-isometric embedding $g : Y \to X$, so there is some constant C such that for all $y \in Y$, $\rho(f \circ g(y), y) \leq c$; in particular, f has quasi-dense image (choose $g(y)$ for the particular choice of x).

Now, suppose f is a quasi-isometric embedding with quasi-dense image. We will construct a quasi-inverse quasi-isometric embedding. We may find a constant C such that for all $x, x' \in X$, we have

$$\frac{1}{C} d(x, x') - C \leq \rho(f(x), f(x')) \leq C d(x, x') + C$$

and for all $y \in Y$, there exists $x \in X$ such that

$$\rho(f(x), y) \leq C.$$

For each y, we may then choose x_y such that the map $g\colon Y \to X$ taking $y \mapsto x_y$ has $d\Big(f\Big(x_y\Big), y\Big) \leq C$ holding for all $y \in Y$.

We claim that the map g is quasi-inverse to f. We observe that by definition, we have

$$\begin{aligned}\rho(f \circ g(y), y) &= \rho\Big(f\Big(x_y\Big), y\Big)\\ &\leq C.\end{aligned}$$

Conversely, using the fact that f is a quasi-isometric embedding, we have for all $x \in X$ that

$$\begin{aligned}d(g \circ f(x), x) &= d\Big(x_{f(x)}, x\Big)\\ &\leq C\rho\Big(f\Big(x_{f(x)}\Big), f(x)\Big) + C^2\\ &\leq 2C^2,\end{aligned}$$

meaning that both $f \circ g$ and $g \circ f$ have finite distance from the respective identity maps.

Finally, we will show that g is a quasi-isometric embedding. A slight modification of this argument will show that quasi-inverses of quasi-isometric embeddings are quasi-isometric embeddings.

$$\begin{aligned}d(g(y), g(y')) &= d\Big(x_y, x_{y'}\Big)\\ &\leq C\rho\Big(f\Big(x_y\Big), f\Big(x_{y'}\Big)\Big) + C^2\\ &\leq C\Big(\rho\Big(f\Big(x_y\Big), y\Big) + \rho(y, y') + \rho\Big(f\Big(x_{y'}\Big), y'\Big)\Big) + C^2\\ &\leq C(\rho(y, y') + 2C) + C^2\\ &= C\rho(y, y') + 3C^2\\ d(g(y), g(y')) &= d\Big(x_y, x_{y'}\Big)\\ &\geq \frac{1}{C}\rho\Big(f\Big(x_y\Big), f\Big(x_{y'}\Big)\Big) - c^2\\ &\geq \frac{1}{C}\Big(\rho(y, y') - \rho\Big(f\Big(x_y\Big), y\Big) - \rho\Big(f_{x_{y'}}, y'\Big)\Big) - c^2\\ &\geq \frac{1}{C}\rho(y, y') - \Big(2 + c^2\Big).\end{aligned}$$

□

The following inheritance properties can be computed directly from their respective definitions.

- Every map at a finite distance from a quasi-isometric embedding (respectively, quasi-isometry) is a quasi-isometric embedding (respectively, quasi-isometry).
- If $f, f'\colon X \to Y$ are maps that are finite distance from each other, then if $g\colon Z \to X$ is any map, $f \circ g$ and $f' \circ g$ have finite distance from each other. If $g\colon Y \to Z$ is a quasi-isometric embedding, then $g \circ f$ and $g \circ f'$ also have finite distance from each other.
- Compositions of quasi-isometric (respectively, bilipschitz) embeddings are quasi-isometric (respectively, bilipschitz) embeddings.
- Compositions of quasi-isometries (respectively, bilipschitz equivalences) are quasi-isometries (respectively, bilipschitz equivalences).

Given a metric space X, the quasi-isometry group of X, denoted $\mathrm{QI}(X)$, is the set of all quasi-isometries $f\colon X \to X$ modulo finite distance, equipped with the composition operation.

Quasi-Isometry of Groups

Definition: Let $X = (V, E)$ be a connected graph. The distance between $v, w \in V$ is equal to the length of the smallest path between v and w. This is often known as the graph metric. Note that it is a metric on the vertex sets.

It can be observed that an isometric isomorphism of vertex sets equipped with this metric is equivalent to an isometric isomorphism of the underlying graphs.

Definition: Let G be a finitely generated group with generating set S. The *word length* of $g \in G$ with respect to S, denoted $\ell_S(g)$, is the smallest $n \geq 0$ such that g may be written as a product of n elements of S. That is,

$$\ell_S(g) = \min\{n : g = s_1 s_2 \cdots s_n \text{ with } s_i \in S \text{ and } n \geq 0\}.$$

The *word metric* is then

$$d_S(g, h) = \ell_S\left(g^{-1}h\right).$$

Proposition: The word metric is a left-invariant metric. That is, it satisfies the following:

(i) $d_S(g, h) = 0$ if and only if $g = h$;

(ii) $d_S(g, h) = d_S(h, g)$;

(iii) $d_S(g, h) \leq d_S(g, k) + d_S(k, h)$;

(iv) $d_S(g, h) = d_S(kg, kh)$.

Proof.

(i) We have that $\ell(g) = 0$ if and only if $g = e$ by the fact that any nontrivial element must be a nontrivial product of elements of S.

(ii) Notice that $\ell(g) = \ell\left(g^{-1}\right)$ since S is symmetric, so if

$$g = s_1 \cdots s_n$$

realizes $\ell(g)$, then

$$g^{-1} = s_n^{-1} \cdots s_1^{-1}$$

must realize $\ell\left(g^{-1}\right)$, as otherwise we would have a smaller representation for g in terms of a product of elements of S.

(iii) Observe that, by the existence of relations and word reductions, we have $\ell(gh) \leq \ell(g) + \ell(h)$.

(iv) We have

$$\begin{aligned} d(kg, kh) &= \ell\left((kg)^{-1}(kh)\right) \\ &= \ell\left(g^{-1}k^{-1}kh\right) \\ &= \ell\left(g^{-1}h\right) \\ &= d(g, h). \end{aligned}$$

□

Proposition: Let G be a finitely generated group, and let S and S' be finite generating sets of G. Then, the identity map is a bilipschitz equivalence of the metric spaces (G, d_S) and $(G, d_{S'})$.

Proof. Since S is finite, the value

$$c = \max\{s \in S\} d_{S'}(e, s)$$

is also finite. Suppose $g, h \in G$, and let $n = d_S(g, h)$. Then, $g^{-1}h = s_1 \cdots s_n$ for some $s_1, \ldots, s_n \in S$. By the triangle inequality and left-invariance of the metric $d_{S'}$, we obtain

$$\begin{aligned} d_{S'}(g, h) &= d_{S'}(g, gs_1 \cdots s_n) \\ &\leq d_{S'}(g, gs_1) + \cdots + d_{S'}(gs_1 \cdots s_{n-1}, gs_1 \cdots s_n) \\ &= d_{S'}(e, s_1) + d_{S'}(e, s_2) + \cdots + d_{S'}(e, s_n) \\ &\leq cn \\ &= c d_S(g, h). \end{aligned}$$

Symmetry gives the desired bilipschitz equivalence. □

Due to the word metric, we are able to define notions of bilipschitz equivalence and quasi-isometry for finitely generated groups.

Definition: Let G be a finitely generated group.

(i) We say G is bilipschitz equivalent to a metric space X if for some finite generating set S, the metric spaces (G, d_S) and X are bilipschitz equivalent.

(ii) We say G is quasi-isometric to a metric space X if for some finite generating set S, the metric spaces (G, d_S) and X are quasi-isometric. We will write $G \sim_{\text{QI}} X$ if G and X are quasi-isometric.

Remark:

(i) Bijective quasi-isometries of finitely generated groups are bilipschitz equivalences due to the fact that the minimum distance between any two group elements is 1.

(ii) If $S \subseteq G$ is a generating set for some group G, then S is finite if and only if the word metric d_S is proper, in the sense that the balls of finite radius in (G, d_S) are finite.

We start by considering the quasi-isometry classification of finite groups. It turns out that a finitely generated group is quasi-isometric to a finite group if and only if it is finite. This follows from the fact that all finite groups yield metric spaces of finite diameter, and all finite diameter metric spaces are quasi-isometric to each other; conversely, a group is quasi-isometric to a finite group if and only if there is a finite generating set that yields a metric space with finite diameter, but since balls are finite, it follows that the group is itself finite.

Next, we claim that the groups $\mathbb{Z}$, $\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, and $D_\infty \cong \mathbb{Z} \rtimes \mathbb{Z}/2\mathbb{Z}$ are bilipschitz equivalent. Consider the equivalent presentations

$$\begin{aligned} D_\infty &= \left\langle x, y | x^2, y^2 \right\rangle \\ &= \left\langle s, t | t^2, tst^{-1} = s^{-1} \right\rangle. \end{aligned}$$

Then, the Cayley graph $\text{Cay}(D_\infty, \{x, y\})$ is isomorphic to the Cayley graph $\text{Cay}(\mathbb{Z}, \{1\})$ — map $\{x, xy, xyx, \ldots\}$ to strictly positive numbers and map $\{y, yx, yxy, \ldots\}$ to strictly negative numbers.

On the other hand, the Cayley graph $\text{Cay}(D_\infty, \{s, t\})$ is isomorphic to the Cayley graph $\text{Cay}(\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}, \{(1, [0]), (0, [1])\})$. Since the word metrics on D_∞ with respect to $\{x, y\}$ and $\{s, t\}$ are bilipschitz equivalent, we get our desired result.

In fact, it is possible to show something a lot more rigid.

Theorem: A finitely generated group is quasi-isometric to $\mathbb{Z}$ if and only if there is a finite-index subgroup isomorphic to $\mathbb{Z}$.

Quasi-Geodesics

Definition: Let (X,d) be a metric space.

(i) A geodesic of length $L \geq 0$ is an isometric embedding $\gamma\colon [0,L] \to X$, where $[0,L]$ is endowed with the standard metric on $\mathbb{R}$. The point $\gamma(0)$ is the start point of γ and the point $\gamma(L)$ is the endpoint of γ.

(ii) The metric space X is called geodesic if for all $x, x' \in X$ there exists a geodesic in X with start point x and end point x'.

(iii) Given $c > 0$ and $b \geq 0$, a (c,b)-quasi-geodesic is a (c,b)-quasi-isometric embedding $\gamma\colon [t,t'] \to X$. The point $\gamma(t)$ is the start point of γ and the point $\gamma(t')$ is the end point of γ.

(iv) The space (X,d) is called (c,b)-quasi-geodesic if for all $x, x' \in X$, there exists a (c,b)-quasi-geodesic with start point x and end point x'.

More concretely, for any $x, x' \in X$, there is a map $\gamma\colon [t,t'] \to X$ such that for any $s, s' \in [t,t']$, we have

$$\frac{1}{c}\left|s-s'\right| - b \leq d(\gamma(s),\gamma(s')) \leq c\left|s-s'\right| + b.$$

The most important example of a quasi-geodesic space is that of a connected graph. Specifically, if $X = (V,E)$ is a connected graph, then the graph metric on V (defined as the shortest path between two vertices) turns V into a $(1,1)$-quasi-geodesic space.

It is possible to turn a graph into a geodesic space by gluing any two vertices connected by an edge using a unit interval. In fact, this applies to any quasi-geodesic metric space, and the resulting geodesic space is quasi-isometric to the original quasi-geodesic space.

Milnor–Schwarz Lemma

We wish to understand what finitely-generated groups look like up to quasi-isometry. The Milnor–Schwarz lemma helps establish this via a sufficiently nice action of the group.

Proposition (Milnor–Schwarz Lemma)**:** Let G be a group, and let G act on a nonempty metric space (X,d) by isometries.

Suppose there are constants $c, b > 0$ such that X is (c,b)-quasi-geodesic, and suppose there is a subset $B \subseteq X$ with:

- $\operatorname{diam}(B)$ finite;
- $\bigcup_{g\in G} g.B = X$;
- the action of G satisfies $|\{g \in G : g.B' \cap B' \neq \emptyset\}|$ is finite, where we define

$$B' := \{x \in X : \text{ there exists } y \in B \text{ such that } d(x,y) \leq 2b\}.$$

Then, the following hold.

(i) The group G is generated by S, and in particular, G is finitely generated.

(ii) For all $x \in X$, the map $g \mapsto g.x$ yields a quasi-isometry with respect to the word metric d_S on G.

Proof. We start by showing that S generates G. Let $g \in G$, and let $x \in B$. Since X is (c,b)-quasi-geodesic, there is a (c,b)-quasi-geodesic map $\gamma\colon [0,L] \to X$ from x to $g.x$.

Let $n = \lceil Lc/b \rceil$. For each $j \in \{0,\dots,n-1\}$, we define $t_j = j\frac{b}{c}$, $t_n = L$, and $x_j = \gamma\left(t_j\right)$. Notice that $x_0 = x$ and $x_n = g.x$. Since the translates of B cover all of X, there are group elements $g_j \in G$ with $x_j \in g_j.B$. We may choose $g_0 = e$ and $g_n = g$.

Then, for each $j = 1, \dots, n$, the element $s_j = g_{j-1}^{-1} g_j$ lies in S. Since γ is a (c, b)-quasi-geodesic, it follows that

$$\begin{aligned} d\left(x_{j-1}, x_j\right) &\leq c\left|t_{j-1} - t_j\right| + b \\ &\leq c\frac{b}{c} + b \\ &\leq 2b. \end{aligned}$$

In particular, $x_j \in g_{j-1}.B'$, while the definition of x_j has $x_j \in g_j.B \subseteq g_j.B'$, whence $g_{j-1}.B' \cap g_j.B' \neq \emptyset$. That is, it follows that $s_j = g_{j-1}^{-1} g_j \in S$.

Therefore, we have

$$\begin{aligned} g &= g_n \\ &= g_{n-1} g_{n-1}^{-1} g_n \\ &= g_{n-1} s_{n-1} g_n \\ &\vdots \\ &= s_1 \cdots s_n \end{aligned}$$

is contained in the subgroup generated by S.

Now, we will show that G is quasi-isometric to X. We will show that $g \overset{\varphi}{\mapsto} g.x$ is a quasi-isometric embedding with quasi-dense image. Since G acts by isometries on X and the G-translates of B cover all of X, it follows that we may assume that $x \in B$.

First, we will show that $g \mapsto g.x$ has quasi-dense image. Notice that if $x' \in X$, then there is some $g \in G$ with $x' \in g.B$, so since $g.x \in g.B$, we have

$$\begin{aligned} d(x', \varphi(g)) &= d(x', g.x) \\ &\leq \operatorname{diam}(g.B) \\ &= \operatorname{diam}(B). \end{aligned}$$

Since we assumed that $\operatorname{diam}(B)$ is finite, this yields a quasi-dense image.

Now, we will show that φ is a quasi-isometric embedding. Let $g \in G$ be given by $g = s_1 \cdots s_n$ realizing the word length of g (i.e., the distance to the identity). We will establish a uniform lower bound on $d(\varphi(e), \varphi(g))$ in terms of $d_S(e, g)$. Let $\gamma \colon [0, L] \to X$ be a (c, b)-quasi-geodesic from x to $g.x$. Then, we have

$$\begin{aligned} d(\varphi(e), \varphi(g)) &= d(x, g.x) \\ &= d(\gamma(0), \gamma(L)) \\ &\geq \frac{1}{c} L - b \\ &\geq \frac{1}{c} \frac{b(n-1)}{c} - b \\ &= \frac{b}{c^2} n - \frac{b}{c^2} b \\ &\geq \frac{b}{c^2} d_S(e, g) - \frac{b}{c^2} - b. \end{aligned}$$

Conversely, we use the triangle inequality, the fact that G acts via isometries, and that $s_j.B' \cap B' \neq \emptyset$ to obtain

$$\begin{aligned} d(\varphi(e), \varphi(g)) &= d(x, g.x) \\ &\leq d(x, s_1.x) + \cdots + d((s_1 \cdots s_{n-1}).x, g.x) \end{aligned}$$

$$\begin{aligned} &= d(x, s_1.x) + \cdots + d(x, s_n.x) \\ &\leq 2n(\operatorname{diam}(B) + 2b) \\ &= 2(\operatorname{diam}(B) + 2b) d_S(e, g). \end{aligned}$$

Notice that $d(\varphi(g), \varphi(h)) = d\left(\varphi(e), \varphi\left(g^{-1}h\right)\right)$ and $d_S(g,h) = d_S\left(e, g^{-1}h\right)$ holds for all $g, h \in G$, so that φ is thus a quasi-isometric embedding. □

We next establish a more topological (and useful) formulation of the Milnor–Schwarz Lemma.

Corollary (Topological Formulation of Milnor–Schwarz)**:** Let G be a group acting by isometries on a nonempty proper geodesic metric space (X, d). Suppose this action is proper and cocompact. Then, G is finitely generated, and for all $x \in X$, the map $g \mapsto g.x$ is a quasi-isometry. That is, G and X are quasi-isometric.

First, we recall/define the most important ideas from this formulation.

- A metric space X is called proper if for all $x \in X$ and all $r > 0$, the closed ball $B(x, r)$ is compact with respect to the metric topology.
- An action $G \curvearrowright X$ is called proper if for all compact $B \subseteq X$, the set $\{g \in G : g.B \cap B \neq \emptyset\}$ is finite.
- An action $G \curvearrowright X$ is called cocompact if the (left) quotient space $G\backslash X$ is compact with respect to the quotient topology.

Example:

- The translation action of $\mathbb{Z}$ on $\mathbb{R}$ is proper. Furthermore, this action is cocompact since the quotient is given by S^1
- The action by deck transformations of the fundamental group of a locally compact path-connected topological space on the universal cover is proper. If X is compact, then the action by deck transformations is also cocompact since the quotient is given by $\pi_1(X, x_0)\backslash \widetilde{X} \cong X$.
- Stabilizer groups of a proper action are finite, but the converse is not necessarily true. For instance, the action of $\mathbb{Z}$ on S^1 by irrational rotation is free but not proper.
- The translation action of $\mathbb{Z}$ on $\mathbb{R}^2$ is not cocompact since the quotient is given by $S^1 \times \mathbb{R}$.
- The action of $\mathrm{SL}_2(\mathbb{Z})$ by Möbius transformations on the upper half-plane $\mathbb{H}$ is not cocompact.

Proof of Topological Formulation. Since X is geodesic, it follows that X is $(1, b)$-quasi-geodesic for all $b > 0$. Our goal is to find a suitable subset $B \subseteq X$ that allows us to apply the original Milnor–Schwarz Lemma.

Since the projection $\pi : X \to G\backslash X$ is an open map, and $G\backslash X$ is compact, it follows that there is a subspace $B \subseteq X$ such that $\pi(B) = G\backslash X$. Thus, we have $\bigcup_{g \in G} g.B = X$, and

$$B' = \{y \in X : d(x, y) \leq 2b \text{ for some } x \in B\}$$

has finite diameter. Since X is a proper space, the subset B' is compact. Since the action of G on X is proper, it follows that $\{g \in G : g.B' \cap B' \neq \emptyset\}$ is finite. Thus, the conditions for the Milnor–Schwarz lemma apply. □

Applications of the Milnor–Schwarz Lemma

Corollary: Finite index subgroups of finitely generated groups are finitely generated and quasi-isometric to the ambient group.

Proof. Let $H \subseteq G$ be a finite index subgroup of the finitely generated group G. The left translation action of H on (G, d_S) for some finite generating set S satisfies the conditions of the Milnor–Schwarz lemma.

- The action of H on (G, d_S) is isometric.
- The space (G, d_S) is $(1, 1)$-quasi-geodesic.

- If $B \subseteq G$ is a finite set of representatives for the right cosets $H\backslash G$, then $HB = G$, and

$$B' = \{y \in G : d_S(x, y) \leq 2 \text{ for some } x \in B\}$$

is finite, and the set

$$T = \{h \in H : h.B' \cap B' \neq \emptyset\}$$

is also finite.

Thus, H is finitely generated by T and the inclusion $H \hookrightarrow G$ is a quasi-isometry with respect to any word metric. □

This yields a very significant notion of equivalence between groups.

Definition:

- Two groups G and H are called commensurable if they contain finite index subgroups $G' \subseteq G$ and $H' \subseteq H$ with $G' \cong H'$.
- Two subgroups G and H are weakly commensurable if they contain finite index subgroups $G' \subseteq G$ and $H' \subseteq H$ such that there exist finite normal subgroups $N \trianglelefteq G'$ and $M \trianglelefteq H'$ with isomorphic quotients.

Corollary: Let G be a group.

(i) If $G' \subseteq G$ is a finite index subgroup of G, then G' is finitely generated if and only if G is finitely generated.

(ii) Let N be a finite normal subgroup. Then, G/N is finitely generated if and only if G is finitely generated, and $G \sim_{\text{QI}} G/N$.

Proof.

(i) Using the first corollary to Milnor–Schwarz, it suffices to show that G is finitely generated if G' is. Yet, since $G' \subseteq G$ is finite index, it follows that taking the representatives of the finitely many cosets as well as the generators of G' yields a generating set for G.

(ii) If G is finitely generated, then so is G/N. Conversely, if G/N is finitely generated, then taking representatives $s' \in \pi^{-1}(sN)$ for each sN in the generating set for G/N, along with the finite set N, yields a generating set for G.

If G and G/N are finitely generated, and S is a finite generating set for G/N, then on $(G/N, d_S)$, we may define an isometric action of G by taking $g.x \coloneqq \pi(g)x$ for any $x \in G/N$. This action satisfies the conditions of the Milnor–Schwarz lemma, so that $G \sim_{\text{QI}} G/N$.

□

In particular, this means that if two groups are weakly commensurable, then they're quasi-isometric. However, this doesn't go in reverse.

For instance, if F_3 denotes the free group of rank 3 and F_4 the free group of rank 4, then it can be shown that the finitely generated groups $(F_3 \times F_3) * F_3$ and $(F_3 \times F_3) * F_4$ are bilipschitz equivalent, but they are not commensurable since the Euler characteristic of the former is zero while the Euler characteristic of the latter is nonzero.

Now, we will discuss some applications of the Milnor–Schwarz lemma to Riemannian geometry.

Corollary: Let M be a closed connected Riemannian manifold with Riemannian universal cover $\widetilde{M}$. Then, $\pi_1(M)$ is finitely generated and the action of $\pi_1(M)$ on $\widetilde{M}$ by deck transformations yields a quasi-isometry between $\pi_1(M)$ and $\widetilde{M}$.

Proof Sketch. First, one uses arguments from Riemannian geometry to show that $\widetilde{M}$ is a proper geodesic metric space, and the action of $\pi_1(M)$ on $\widetilde{M}$ is isometric, proper (fundamental groups act properly on the universal cover by deck transformations), and cocompact (since the quotient is given by the compact space M).

Therefore, the topological formulation of the Milnor–Schwarz lemma yields the final proof. □

Definition: We call a Riemannian manifold flat if its Riemannian universal cover is isometrically isomorphic to the Euclidean space $\mathbb{R}^n$ of the same dimension.

We call a Riemannian manifold hyperbolic if its Riemannian universal cover is isometrically isomorphic to the hyperbolic space $\mathbb{H}^n$ of the same dimension.

These definitions allow us to yield the following classification.

Corollary:

(i) If M is a closed connected flat Riemannian n-manifold, then its fundamental group is quasi-isometric to $\mathbb{R}^n$, hence quasi-isometric to $\mathbb{Z}^n$.

(ii) If the fundamental group of a closed connected smooth n-manifold is not quasi-isometric to $\mathbb{R}^n$ or $\mathbb{Z}^n$, then the manifold does not admit a flat Riemannian metric.

(iii) If M is a closed connected hyperbolic Riemannian manifold, then $\pi_1(M)$ is quasi-isometric to the hyperbolic space $\mathbb{H}^n$.

(iv) If the fundamental group of a closed connected smooth n-manifold is not quasi-isometric to $\mathbb{H}^n$, then this manifold does not admit a hyperbolic Riemannian metric.

Dynamic Criterion for Quasi-Isometry

Now, we discuss the dynamic criterion for quasi-isometry.

Definition: Let G and H be groups. A set-theoretic coupling for G and H is a nonempty set X with a left action $G \curvearrowright X$ and right action $X \curvearrowleft H$ that commute with each other, in that $(g.x).h = g.(x.h)$, such that X admits a subset K satisfying the following.

- The G and H translates of K cover K. That is, $G.K = K.H$.
- The sets

$$F_G = \{g \in G : g.K \cap K \neq \emptyset\}$$
$$F_H = \{h \in H : K.h \cap H \neq \emptyset\}$$

  are finite.
- For each $g \in G$, there is a finite subset $F_H(g) \subseteq H$ with $g.K \subseteq K.F_H(g)$, and for each $h \in H$, there is a finite subset $F_G(h) \subseteq H$ such that $K.h \subseteq F_G(h).K$.

For instance, if $G, H \subseteq X$ are finite-index subgroups acting by multiplication on the left and right respectively, then G and H admit a set-theoretic coupling since multiplication in X is associative and the subset K can be given by taking the finite sets of representatives for the right cosets of G and left cosets of H.[I]

Definition: Let G and H be groups. A topological coupling for G and H is a nonempty locally compact space X with a proper cocompact left action $G \curvearrowright X$ by homeomorphisms and a proper cocompact right action $X \curvearrowleft H$ by homeomorphisms that commute with each other.

Theorem (Gromov's Coupling Criterion for Quasi-Isometry)**:** Let G and H be finitely generated groups. The following are equivalent:

(i) there is a topological coupling for G and H;

(ii) there is a set-theoretic coupling for G and H;

(iii) the groups G and H are quasi-isometric.

Proof. First, assuming (i), we observe that there is a compact subset $K \subseteq X$ such that $G.K = X = K.H$. Since

[I] The G-orbit of an element $x \in X$ is given by Gx and the H-orbit of x is given by xH.

the actions of G and H are proper, the sets

$$F_G = \{g \in G : g.K \cap K \neq \emptyset\}$$
$$F_H = \{h \in H : K.h \cap K \neq \emptyset\}$$

are finite. Compactness of K and the local compactness of X yield that for every $g \in G$, there is a finite subset $F_H(g) \subseteq H$ such that $g.K \subseteq K.F_H(g)$, and similarly for elements of H. Thus, we obtain a set-theoretic coupling.

Now, we will show that if G and H admit a set-theoretic coupling, G and H are quasi-isometric. Given $x \in K \subseteq X$, we obtain a map $f : G \to H$ such that $g^{-1}.x \in K.f(g)^{-1}$ for all $g \in G$. Given finite generating sets $S \subseteq G$ and $T \subseteq H$, then for any subsets $A \subseteq G$ and $B \subseteq H$, we define

$$D_T B \coloneqq \sup_{b \in B} d_T(e, b)$$
$$D_S A \coloneqq \sup_{a \in A} d_S(e, a).$$

First, we show that the map f has quasi-dense image. Let $h \in H$. Since $G.K = X$, we will find some $g \in G$ with $x.h \in g.K$. Since the actions of G and H commute, we have $g^{-1}.x \in K.h^{-1}$. Yet, by the definition of f, $g^{-1}.x \in K.f(g)^{-1}$, so $K.h^{-1} \cap K.f(g)^{-1} \neq \emptyset$, meaning $h^{-1}f(g) \in F_H$. In particular, this means $d_T(h, f(g)) \leq D_T F_H$, where we recall that F_H is the set of all $h \in H$ such that $K.h \cap K \neq \emptyset$.

Now, we will show that f is a quasi-isometric embedding. Let

$$F_H(S) = \bigcup_{s \in S \cup S^{-1}} F_H(s)$$
$$F_H(T) = \bigcup_{t \in T \cup T^{-1}} F_G(t).$$

These sets are assumed finite since each of the $F_G(t)$ and $F_H(s)$ are finite. Let $g, g' \in G$.

- Let $n = d_S(g, g')$. We start by showing that $K.f(g)^{-1}f(g') \cap K.F_H(S)^n$ is nonempty. First, we see that $g^{-1}.x.f(g') \in K.f(g)^{-1}f(g')$ by the construction of f. We may write $g^{-1}g' = s_1 \cdots s_n$ for some $s_1, \ldots, s_n \in S \cup S^{-1}$, so that

$$\begin{aligned} g^{-1}.x.f(g') &= g^{-1}g'.\left(g'^{-1}.x.f(g')\right) \\ &\in g^{-1}g'.K.f(g')^{-1}f(g') \\ &= g^{-1}g'.K \\ &= s_1 \cdots s_{n-1}s_n.K \\ &\subseteq s_1 \cdots s_{n-1}.K.F_H(S) \\ &\vdots \\ &\subseteq K.F_H(S)^n. \end{aligned}$$

  Thus, we have $K.f(g)^{-1}f(g') \cap K.F_H(S)^n \neq \emptyset$. Notice that all of these calculations use the fact that the actions between G and H commute. By the definition of F_H, we have $f(g)^{-1}f(g') \in F_H F_H(S)^n$. By the triangle inequality, we have

$$\begin{aligned} d_G(f(g), f(g')) &= d_G\left(e, f(g)^{-1}f(g')\right) \\ &\leq D_T(F_H F_H(S)^n) \\ &\leq nD_T F_H(S) + D_T F_H \\ &= d_S(g, g')D_T F_H(S) + D_T F_H. \end{aligned}$$

  This gives us our desired upper bound.

- Using a similar line of reasoning, if $m = D_T(f(g), f(g'))$, then we may find that $g^{-1}.x.f(g') \in F_G(T)^m.K \cap g^{-1}g'.K$. Therefore, this intersection is nonempty, so we may conclude that

$$d_S(g,g') \leq D_S F_G(T) d_T(f(g), f(g')) + D_S F_G.$$

This gives our desired lower bound.

Finally, suppose G and H are quasi-isometric. We will determine a locally convex space that allows for a topological coupling.

Let $S \subseteq G$ and $T \subseteq H$ be finite generating sets for G and H. We will show that there is a finite group F and a constant $C > 0$ such that the set X of all functions $f : G \to H \times F$ satisfying

- f has C-dense image in $H \times F$
- for all $g, g' \in G$,

$$\frac{1}{C} d_S(g,g') \leq d_{T\times F}(f(g), f(g')) \leq C d_S(g,g')$$

is nonempty. Let $f : G \to H$ be a quasi-isometry. Since f is a quasi-isometry, there is some $c > 0$ such that f has c-dense image in H — that is, for all $g, g' \in G$,

$$\frac{1}{c} d_S(g,g') - c \leq d_T(f(g), f(g')) \leq c d_S(g,g') + c.$$

Notice that if $g, g' \in G$ satisfy $f(g) = f(g')$, then $d_S(g,g') \leq c^2$. If F is a finite group that has more elements than $B_S\left(e_G, c^2\right)$, then we may construct an injective quasi-isometry $\overline{f} : G \to H \times F$. Let $\overline{c} > 0$ be such that $\overline{f}$ is a $(\overline{c}, \overline{c})$-quasi-isometric embedding with $\overline{c}$-dense image.

Since $\overline{f}$ is injective, it follows that $\overline{f}$ satisfies a $\max\left(2\overline{c}^2 + \overline{c}\right)$-bilipschitz estimate. Therefore, if we let $C = \max\left(2\overline{c}, \overline{c}^2 + \overline{c}\right)$, the group F and the constant C satisfy the desired properties, whence X is nonempty.

Let G act on X by $(g.f)(x) = f\left(g^{-1}.x\right)$ and H on X by $(f.h)(x) = f(x)(h, e)$. These actions necessarily commute with each other by the way they're defined.

Equipping X with the topology of pointwise convergence, the Arzelà–Ascoli theorem yields that X is locally compact with respect to this topology. This follows from the fact that the elements of X have a bilipschitz condition attached to them so that the family is equicontinuous. Using the Arzelà–Ascoli theorem on the actions of G and H on X yields that the actions are proper and cocompact. □

The most important use case for the dynamic criterion is the consideration of discrete subgroups of locally compact groups.

Definition: Let G be a locally compact topological group. A *uniform lattice* in G is a discrete subgroup Γ of G such that the left translation action (or right translation action) of Γ on G is cocompact.

Corollary: Let G be a locally compact topological group. Then, all finitely generated uniform lattices in G are quasi-isometric.

Proof. Let Γ and Λ be finitely generated uniform lattices in G. Then, the left and right translation actions of Γ and Λ respectively are continuous (by definition) and commute (since multiplication is associative).

These actions are cocompact and proper, so G is a topological coupling for Γ and Λ, whence Γ and Λ are quasi-isometric. □

The best use of the dynamic criterion is that one can apply it to other settings. The measure-theoretic notion is known as measure equivalence, and it plays an essential role in measured group theory.

Now, we will discuss some other quasi-isometry invariants.

Asymptotic Cones

Let (X,d) be a metric space, and let $x_0 \in X$ be the base point. Let ω be a free ultrafilter on $\mathbb{N}$.

Definition: We say a sequence $(x_n)_n = x$ is *moderate* if there exists some constant A (dependent on x) such that

$$d(x_n, x_0) \leq An$$

for all $n \in \mathbb{N}$. Let $\mathcal{M}(X,d)$ be the set of all moderate sequences.

That the sequences in $\mathcal{M}(X,d)$ are invariant under basepoint follows from the fact that, whenever $n \geq 1$, we have $An + C \leq (A+C)n$.

If $x, y \in X^{\mathbb{N}}$ are moderate sequences, we observe that

$$\begin{aligned} d(x_n, y_n) &\leq d(x_n, x_0) + d(x_0, y_n) \\ &\leq \Big(A_x + A_y\Big)n. \end{aligned}$$

This gives that the sequence $\Big(n^{-1} d(x_n, y_n)\Big)_n$ is bounded, so that the ultrafilter limit

$$\lim_\omega \frac{d(x_n, y_n)}{n}$$

always exists. Write $x \sim_\omega y$ if $\lim_\omega n^{-1} d(x_n, y_n) = 0$. This is an equivalence relation on $\mathcal{M}(X,d)$. Write

$$K_\omega(X,d) = \mathcal{M}(X,d)/\sim_\omega$$

for the quotient space. We have a well-defined distance metric given by

$$d_\omega\Big([x]_\omega, [y]_\omega\Big) := \lim_\omega \frac{d(x_n, y_n)}{n}.$$

We call the metric space $(K_\omega(X,d), d_\omega)$ the *asymptotic cone* of the metric space (X,d) relative to the free ultrafilter ω.

The main asymptotic cone we will be concerned with is the asymptotic cone of a metric group (such as a finitely generated group with the word metric).

Example: Let (G,d) be a group generated by a finite symmetric subset $S \subseteq G$ with the metric $d = d_S$. Let e_G be the basepoint for this metric space.

Equip $\mathcal{M}(G,d)$ with the group structure given by pointwise multiplication — that is, $gh = (g_n h_n)_n$. Left invariance gives that

$$\begin{aligned} d(g_n h_n, e_G) &\leq d(g_n, e_G) + d(h_n, e_G) \\ &\leq \Big(A_g + A_h\Big)n. \end{aligned}$$

The constant sequence $e = (e_G)_n$ is the neutral element, and inverses are given by $g^{-1} = \Big(g_n^{-1}\Big)_n$. If ω is a free ultrafilter, and

$$N_\omega = \left\{ g \in \mathcal{M}(G,d) : \lim_\omega n^{-1} d(g_n, e_G) = 0 \right\},$$

then N_ω is a subgroup of $\mathcal{M}(G,d)$, and is such that $g \sim_\omega g'$ if and only if $gN_\omega = g'n_\omega$ for all $g, g' \in \mathcal{M}(G,d)$. There is then a bijection between the asymptotic cone and the coset space of N_ω in $\mathcal{M}(G,d)$. That is,

$$K_\omega(G,d) \cong \frac{\mathcal{M}(G,d)}{N_\omega}.$$

Furthermore, if d is bi-invariant, then N_ω is normal and $K_\omega(G,d)$ equipped with the metric d_ω is a metric group.

Asymptotic cones are the structure that allow us to convert the information in quasi-isometries into bilipschitz equivalences. In a sense, asymptotic cones allow us to "zoom out" of a metric space in a well-defined fashion so that we may capture the coarse structure of a quasi-isometry.

Proposition: Suppose (X,d) and (Y,d) are metric spaces, and ω is a free ultrafilter on $\mathbb{N}$. Every quasi-isometry $\Phi\colon X\to Y$ induces a bilipschitz homeomorphism $\Phi_\omega\colon K_\omega(X,d)\to K_\omega(Y,\rho)$.

Proof. Let $C\geq 1$ be sufficiently large such that Φ satisfies

$$\frac{1}{C}d(x_1,x_2)-C\leq \rho(\Phi(x_1),\Phi(x_2))\leq Cd(x_1,x_2)+C$$

and is C-quasi-dense. That is, for all $x\in X$, there is $y\in Y$ such that

$$\rho(\Phi(x),y)\leq C.$$

Let x_0 be a basepoint for X, and let $y_0=\Phi(x_0)$ be the corresponding basepoint for Y. Then, for any $x\in \mathcal{M}(X,d)$, we have

$$\begin{aligned}\rho(\Phi(x_n),y_0)&\leq Cd(x_n,x_0)+C\\&\leq C(A_x+1)n.\end{aligned}$$

Therefore, $(\Phi(x_n))_{n\in\mathbb{N}}$ is a moderate sequence in Y. We have

$$\begin{aligned}\lim_\omega \frac{\rho(\Phi(x_n),\Phi(x_n'))}{n}&\leq \lim_\omega \frac{Cd(x_n,x_n')+C}{n}\\&=Cd_\omega([x]_\omega,[x']_\omega).\end{aligned}$$

Therefore, the map Φ_ω given by $[x]_\omega\mapsto\left[(\Phi(x_n))_n\right]_\omega$ is well-defined. Furthermore, we have

$$\rho_\omega(\Phi_\omega([x]_\omega),\Phi_\omega([x']_\omega))\leq Cd_\omega([x]_\omega,[x']_\omega).$$

Similarly, we obtain the lower bound given by

$$\begin{aligned}\lim_\omega \frac{d(\Phi_\omega([x]_\omega),\Phi_\omega([x']_\omega))}{n}&\geq \lim_\omega \frac{\frac{1}{C}d(x_n,x_n')+C}{n}\\&=\frac{1}{C}d_\omega([x]_\omega,[x']_\omega).\end{aligned}$$

Finally, we only need to show that Φ_ω is surjective. Let $y\in\mathcal{M}(X,d)$. Then, for each $n\in\mathbb{N}$, there is $x_n\in X$ such that $\rho(\Phi(x_n),y_n)\leq C$. We see that the sequence $x=(x_n)_n$ satisfies

$$\begin{aligned}d(x_n,x_0)&\leq C\rho(\Phi(x_n,),\Phi(x_0))\\&\leq C(\rho(\Phi(x_n),y_n)+\rho(y_n,y_0))\\&\leq C\Big(C+A_y\Big)n,\end{aligned}$$

so it is moderate. Therefore, we have

$$\begin{aligned}\rho_\omega\Big(\Phi_\omega([x]_\omega),[y]_\omega\Big)&=\lim_\omega\frac{\rho(\Phi(x_n),y_n)}{n}\\&\leq\lim_\omega\frac{C}{n}\\&=0.\end{aligned}$$

□

Growth Rates and Types

Definition: Let G be a finitely generated group, and $S \subseteq G$ a finite symmetric generating set for G. The growth function $\beta_{G,S} : \mathbb{N} \to \mathbb{N}$ is given by

$$r \mapsto \left|B_{G,S}(e,r)\right|,$$

where

$$B_{G,S}(e,r) = \{g \in G : d_S(g,e) \leq r\}.$$

Note that growth functions are dependent on the choice of generating sets. For example, $\beta_{\mathbb{Z},\{1\}}$ is given by $r \mapsto 2r+1$, while $\beta_{\mathbb{Z},\{2,3\}}$ is given by

$$r \mapsto \begin{cases} 1 & r = 0 \\ 5 & r = 1 \\ 6r+1 & r > 1. \end{cases}$$

Lemma: Let G be a finitely generated group, $S \subseteq G$ a finite generating set.

(i) For all $r, r' \in \mathbb{N}$, we have

$$\beta_{G,S}(r+r') \leq \beta_{G,S}(r)\beta_{G,S}(r').$$

(ii) Let G be infinite. Then, $\beta_{G,S}$ is strictly increasing, and satisfies $\beta_{G,S}(r) \geq r$ for all $r \in \mathbb{N}$.

(iii) For all $r \in \mathbb{N}$, we have $\beta_{G,S}(r) \leq \beta_{F(S),S}(r)$.

Proof.

(i) We start by noting that if $g \in B_{G,S}(r+r')$, then we may write $g = s_1 \cdots s_r \cdots s_{r+r'}$, meaning that $g \in B_{G,S}(r)B_{G,S}(r')$, while for any elements $g \in B_{G,S}(r)$ and $h \in B_{G,S}(r')$, we have $d(e,gh) \leq d\left(e,g^{-1}\right)+d(e,h)$, whence $gh \in B_{G,S}(r+r')$. Therefore, we have

$$\begin{aligned} \left|B_{G,S}(r+r')\right| &= \left|B_{G,S}(r)B_{G,S}(r')\right| \\ &\leq \left|B_{G,S}(r)\right|\left|B_{G,S}(r')\right|. \end{aligned}$$

(ii) Since G is infinite, for any $r \in \mathbb{N}$ there exists some $g \in G$ such that $d(e,g) = r$, and furthermore, if $g = s_1 \cdots s_r$, then there is some $s \in S$ such that $d(e,gs) > d(e,g)$, so that $B_{G,S}(e,r) \subsetneq B_{G,S}(e,r+1)$.

(iii) The homomorphism $\varphi\colon F(S) \to G$ given by $\varphi|_S = \mathrm{id}_S$ is contractive with respect to the word metrics given by S on their respective groups (since G possibly admits relations), and φ is surjective. Therefore,

$$\begin{aligned} \beta_{G,S}(r) &= \left|B_{G,S}(e,r)\right| \\ &= \left|\varphi\left(B_{F(S),S}(e,r)\right)\right| \\ &\leq \left|B_{F(S),S}(e,r)\right| \\ &= \beta_{F(S),S}(r). \end{aligned}$$

□

Now that we have established some properties of the growth rate, we can abstract away from the growth rate to focus, in a sense, on the "largest" term.

Definition:

- A generalized growth function is an increasing function from $\mathbb{R}_{\geq 0}$ to $\mathbb{R}_{\geq 0}$.

- If f, g are generalized growth functions, we say that g *quasi-dominates* f if there exist $b, c \in \mathbb{R}_{>0}$ such that

$$f(r) \leq cg(cr + b) + b.$$

  We write $f \preceq g$.
- If $f \preceq g$ and $g \preceq f$, then we say f and g are *quasi-equivalent*, and write $f \sim g$.

Example:

- If $a \geq 0$, then $x \mapsto x^a$ is a generalized growth function, and $(x \mapsto x^a) \preceq \left(x \mapsto x^{a'}\right)$ if and only if $a \leq a'$. Notice that $(x \mapsto x^a) \sim \left(x \mapsto x^{a'}\right)$ if and only if $a = a'$.
- If $a > 1$, then $x \mapsto a^x$ is a generalized growth function. Notice that $x \mapsto a^x$ is quasi-equivalent to $x \mapsto a'^x$ for all $a, a' > 1$, since we may substitute $\log_a(a')x$ in the first equation to yield the second equation, and vice versa.

Notice that if G is a finitely generated group with finite generating set S, then the function $r \mapsto \beta_{G,S}(\lceil r \rceil)$ is a submultiplicative generalized growth function. In particular, $\beta_{G,S}(r) \preceq \beta_{H,T}(r)$ if and only if there exists $b, c \in \mathbb{N}$ such that

$$\beta_{G,S}(r) \leq c\beta_{H,T}(cr + b) + b$$

for all $r \in \mathbb{N}$.

Our primary focus will be on quasi-isometry invariance of the growth rate.

Proposition: Let G and H be finitely generated groups with generating sets $S \subseteq G$ and $T \subseteq H$. If there exists a quasi-isometric embedding $(G, d_S) \to (H, d_T)$, then $\beta_{G,S} \preceq \beta_{H,T}$.

In particular, if G and H are quasi-isometric, then the growth functions $\beta_{G,S}$ and $\beta_{H,T}$ are quasi-equivalent.

Proof. Let $f : G \to H$ be a quasi-isometric embedding. There exists $c > 0$ such that

$$\frac{1}{c} d_S(g, g') - c \leq d_T(f(g), f(g')) \leq c d_S(g, g') + c.$$

Write $e' = f(e)$, and let $r \in \mathbb{N}$. If $g \in B_{G,S}(e, r)$, then we have

$$\begin{aligned} d_T(f(g), e') &\leq c d_S(g, e) + c \\ &\leq cr + c, \end{aligned}$$

so that $f(B_{G,S}(e, r)) \subseteq B_{H,T}(e', cr + c)$. Furthermore, if $f(g) = f(g')$, then

$$\begin{aligned} d_S(g, g') &\leq c(d_T(f(g), f(g')) + c) \\ &= c^2. \end{aligned}$$

Since the metrics are invariant under left translation, we get

$$\begin{aligned} \beta_{G,S}(r) &\leq \left|B_{G,S}\left(e, c^2\right)\right| \left|\beta_{H,T}(e', cr + c)\right| \\ &= \left|B_{G,S}\left(e, c^2\right)\right| \left|B_{H,T}(e, cr + c)\right| \\ &= \beta_{G,S}\left(c^2\right) \beta_{H,T}(cr + c). \end{aligned}$$

Since c is constant, it follows that $\beta_{G,S} \preceq \beta_{H,T}$. □

This allows us a well-defined notion of growth type for a finitely generated group (recall that bilipschitz equivalences are quasi-isometries, so the growth type is invariant with respect to the generating set).

Definition: Let G be a finitely generated group.

(i) The growth type of G is the quasi-equivalence class of a growth function of G with respect to any finite generating set.

(ii) The group G is said to be of *exponential growth* if it is of type $x \mapsto e^x$.

(iii) The group is said to be of *polynomial growth* if it is of type $(x \mapsto x^a)$ for some $a \geq 0$.

(iv) The group is said to be of intermediate growth if it is neither exponential nor polynomial growth.

This trichotomy emerges from the fact that any group has its growth dominated by the growth of the free group on its generating set, which is exponential.

Amenability of Groups

We are only concerned with discrete groups.

Means on Groups

Definition: A group G is called *amenable* if there exists a map $m\colon \ell^\infty(G) \to \mathbb{C}$ (known as a mean) satisfying

- $m(1) = 1$ (that is, the map $1\colon G \to \mathbb{C}$ taking all elements to 1);
- $m(f) \geq 0$ for all positive $f \in \ell^\infty(G)$;
- for all $g \in G$ and all $f \in \ell^\infty(G)$, we have $m(g.f) = m(f)$, where

$$g.(f)(s) = f\left(g^{-1}s\right)$$

is the left regular action of G on itself.

The set of means in $\ell^\infty(G)^*$ is a compact convex subset of the ball $B_{\ell^\infty(G)^*}$. This follows from the fact that if m_1 and m_2 are means, then for any $0 \leq t \leq 1$, $(tm_1 + (1-t)m_2)(1) = 1$, and $(tm_1 + (1-t)m_2)(f) \geq 0$ for all $f \geq 0$. Finally, a net of means admits a subnet in the ball $B_{\ell^\infty(G)^*}$ by Banach–Alaoglu, and this subnet is a mean, so that any net of means admits a subnet converging to a mean.

Definition: Let G be a group. We define the set $\mathrm{Prob}(G)$ to be the set of all left-invariant probability measure on the power set $P(G)$. That is, a measure $\mu\colon P(G) \to [0,1]$ satisfying $\mu(G) = 1$ and $\mu(gA) = \mu(A)$ for all $A \subseteq G$.

Proposition: Given a mean $m \in \ell^\infty(G)^*$, define a probability measure $\hat{m}$ by taking

$$\hat{m}(A) = m(\chi_A).$$

Then, the map Φ taking $m \mapsto \hat{m}$ is a bijection between the set of means on G and the set of invariant probability measures on G.

Proof. Recall that the set of all finitely supported functions $\mathbb{C}[G] \subseteq \ell^\infty(G)$ is norm-dense.

Given a measure $\mu \in \mathrm{Prob}(G)$, define a map $\overline{\mu}\colon \mathbb{C}[G] \to \mathbb{C}$ by

$$\overline{\mu}(f) = \sum_{\lambda \in \mathbb{C}} \mu\left(f^{-1}(\{\lambda\})\right)\lambda.$$

We may extend $\overline{\mu}$ to a map $\widetilde{\mu}\colon \ell^\infty(G) \to \mathbb{C}$, satisfying

$$\begin{aligned}\widetilde{\mu}(1) &= \overline{\mu}(1)\\ &= \mu(G)\\ &= 1.\end{aligned}$$

For any $y \in \mathbb{C}[G]$ with $y \geq 0$, one has $\widetilde{\mu}(y) = \mu(y) \geq 0$. Given any $f \in \ell^\infty(G)$ with $f \geq 0$, we may find a sequence of positive elements $(z_n)_n \to f$. Since $\widetilde{\mu}$ is continuous, it follows that

$$\begin{aligned}\widetilde{\mu}(f) &= \lim_{n\to\infty} \overline{\mu}(z_n) \\ &\geq 0,\end{aligned}$$

so that $\widetilde{\mu}$ is a mean.

Meanwhile, for any $A \subseteq G$, we have $\widetilde{\mu}(\chi_A) = \overline{\mu}(\chi_A) = \mu(A)$, so that Φ is surjective.

To show that Φ is injective, we consider means m_1 and m_2 with $\Phi(m_1) = \Phi(m_2)$. Then, m_1 and m_2 coincide on $\mathbb{C}[G]$, so since $\mathbb{C}[G]$ is dense in $\ell^\infty(G)$, it follows that $m_1 = m_2$ by continuity. Thus, Φ is injective. □

Therefore, we may equivalently define an amenable group to be a group that admits a left-invariant probability measure on $P(G)$.

Examples and Inheritance Properties

Proposition: Finite groups are amenable.

Proof. The averaging operator

$$f \mapsto \frac{1}{|G|} \sum_{g\in G} f(g)$$

is a mean. □

Proposition: Subgroups of amenable groups are amenable.

Proof. Let G be an amenable group, $H \leq G$, and let $\mu\colon P(G) \to [0,1]$ a left-invariant probability measure. Let R be a complete set of representatives for the right cosets G/H. Then, for each $g \in G$, there is a unique $(h, r) \in H \times R$ such that $g = hr$. Defining

$$\widetilde{\mu}(A) = \mu\left(\bigcup_{r\in R} Ar\right)$$

is a left-invariant finitely additive probability measure. □

Proposition: Quotients of amenable groups are amenable.

Proof. Let G be amenable, $N \trianglelefteq G$ a normal subgroup, and let $\rho\colon G \to G/N$ the canonical homomorphism. Define $\hat{\mu}\colon P(G/N) \to [0,1]$ by

$$\hat{\mu}(A) := \mu\Big(\rho^{-1}(A)\Big).$$

□

Proposition: Let G be a group, and let H be a normal subgroup of G. Suppose H and G/H are amenable. Then, G is amenable.

Proof. Find a left-invariant mean $m_0 \in \ell^\infty(H)^*$. Let $f \in \ell^\infty(G)$. Let

$$\widetilde{x}(\overline{g}) = m_0\Big(\big(g^{-1}.x\big)|_H\Big),$$

where $\overline{g} = gH \in G/H$ is the equivalence class of g modulo H. Then, $\widetilde{x}$ is well-defined, since if $\overline{g_1} = \overline{g_2}$, then there is h such that $g_1 = g_2 h$, whence

$$\begin{aligned}m_0\Big(\big(g_2^{-1}.x\big)|_H\Big) &= m_0\Big(h^{-1}.\big(\big(g_1^{-1}.x\big)|_H\big)\Big) \\ &= m_0\Big(\big(g_1^{-1}.x\big)|_H\Big),\end{aligned}$$

where the latter equivalence follows from left invariance of m_0. Furthermore, $\widetilde{x} \in \ell^\infty(G/H)$. Since G/H is amenable, there is a mean $m_1 \in \ell^\infty(G/H)^*$. Set $m(x) = m_1(\widetilde{x})$ for any $x \in \ell^\infty(G)$.

We will show that m is invariant. Let $g \in G$, and let $x \in \ell^\infty(G)$. Then, for all $g' \in G$, we have

$$\begin{aligned}\widetilde{g.x}\left(\overline{g'}\right) &= m_0\left(\left(g'^{-1}g.x\right)|_H\right)\\ &= m_0\left(\left(\left(g^{-1}g'\right)^{-1}.x\right)|_H\right)\\ &= \widetilde{x}\left(\overline{g^{-1}g'}\right)\\ &= \overline{g}.\widetilde{x}\left(\overline{g'}\right).\end{aligned}$$

Therefore, $\widetilde{g.x} = \overline{g}.\widetilde{x}$. Since m_1 is left-invariant under G/H, we get our desired result. □

Corollary: Every virtually amenable group is amenable.

Proof. Let G be a virtually amenable group, and let H be an amenable subgroup of finite index. Then, the set $K = \bigcap_{g\in G} gHg^{-1}$ is a normal subgroup of finite index contained in H. Therefore, K is amenable since it is a subgroup of an amenable group, while G/K is finite (hence amenable), so G is amenable. □

Proposition: Every group that is the inductive limit of a system of amenable groups is amenable.

Proof. Let $(G_i)_i$ be an inductive system of amenable groups, and let $G = \varinjlim G_i$. Let $H_i = h_i(G_i)$, where $h_i \colon G_i \to G$ is the canonical inclusion homomorphism. Since each H_i is amenable, we may find a family $\widetilde{m_i} \in \ell^\infty(H_i)^*$ of means.

Let $m_i \in \ell^\infty(G)^*$ be a family of means defined by

$$m_i(x) = \widetilde{m_i}\left(x|_{H_i}\right).$$

For each $g \in G$, there is some $i_0(g) \in I$ such that $g \in H_i$ for all $i \geq i_0(g)$. It follows that $g.m_i(x) = \widetilde{m_i}\left(g.x|_{H_i}\right) = m_i(x)$ for all $i \geq i_0$; in particular, extracting a subnet, we have that $m = \lim_j m_{i_j}$ is an invariant mean. □

Corollary: Every locally amenable (all finitely generated subgroups are amenable) group is amenable.

Corollary: Every locally finite group is amenable.

Approximation Properties

The next topic of discussion will be by understanding amenability as a type of approximation property. We will need to recall the following two results.

Definition: Let $(m_i)_i \subseteq \ell^1(G)$ be a net in $\ell^1(G)$. We say $(m_i)_i$ is an approximate mean if for all $g \in G$, one has $\|m_i - g.m_i\|_{\ell^1} \to 0$. As always, the left action is defined as $g.m_i(h) = m_i\left(g^{-1}h\right)$.

Theorem: If a group admits an approximate mean, then it admits a mean.

Proof. Let $(m_i)_i$ be an approximate mean. Then, the net of states $(\hat{m}_i)_i$ are contained in $B_{\ell^\infty(G)^*}$. There is then a subnet $\left(\hat{m}_{i_j}\right)_j \to m \in \ell^\infty(G)^*$, where m is a state since the state space is a wk*-closed subset of $B_{\ell^\infty(G)^*}$. Then, m is a mean. □

Theorem: If a group admits a mean, then it admits an approximate mean.

Proof. Suppose G admits a mean, but it doesn't admit an approximate mean. Since finitely supported functions are dense in $\ell^1(G)$, there exists some $\varepsilon_0 > 0$ and some finite subset $E_0 \subseteq G$ such that for all finitely supported $f \in \mathbb{C}[G]$ and all $g \in E_0$, one has $\|f - g.f\|_{\ell^1} \geq \varepsilon_0$.

Consider the normed vector space

$$X = \bigoplus_{|E_0|} \ell^1(G),$$

equipped with the norm

$$\left\|\left(f_g\right)_{g\in E_0}\right\| = \sum_{g\in E_0}\left\|f_g\right\|_{\ell^1}.$$

It follows that

$$C = \left\{\left(f - g.f\right)_{g\in E_0} : f \in \mathbb{C}[G]\right\}$$

is convex with $0 \notin C$. Therefore, by Hahn–Banach separation, we may find some continuous linear functional $\varphi \in X^*$ such that $\mathrm{Re}(\varphi(C)) \subseteq [1,\infty)$. Notice that

$$X^* \cong \bigoplus_{|E_0|} \ell^\infty(G)$$

endowed with the max norm. That is, $\left(u_g\right)_{g\in E_0}$ has norm given by $\max_{g\in E_0}\left\|u_g\right\|_{\ell^\infty}$. Elements $u \in X^*$ are given by $\sum_{g\in G} u_g$ with $u_g \in \ell^\infty(G)$.

Letting $u = \mathrm{Re}(\varphi)$, we see that for all $f \in \mathbb{C}[G]$, we have

$$\begin{aligned} 1 &\leq u\Big(\left(f - g.f\right)_{g\in E_0}\Big) \\ &= \sum_{g\in E_0}\sum_{h\in G} f(h)\Big(u_g(h) - g^{-1}.u_g(h)\Big). \end{aligned}$$

Since this holds for all $f \in \mathbb{C}[G]$, it certainly holds for $\delta_h \in \mathbb{C}[G]$, so that

$$\begin{aligned} 1 &\leq u\Big(\left(\delta_h - g.\delta_h\right)_{g\in E_0}\Big) \\ &= \sum_{g\in E_0}\Big(u_g - g^{-1}.u_g\Big)(h). \end{aligned}$$

Since this holds for all $h \in G$, it follows that, as elements in ℓ^∞, $1 \leq u_g - g^{-1}.u_g$. Applying the mean yields the desired contradiction. □

Definition: Let G be a (finitely generated) group. We say G satisfies the *Følner condition* if, for all finite subsets $E \subseteq G$ and all $\varepsilon > 0$, there exists a nonempty finite subset $F \subseteq G$ with

$$\frac{|gF \triangle F|}{|F|} \leq \varepsilon,$$

or

$$\frac{|gF \cap F|}{|F|} \geq 1 - \varepsilon$$

for all $g \in E$.

We note that, iteratively, one can construct a *Følner sequence* in this fashion, satisfying

$$\left(\frac{|gF_n \triangle F_n|}{|F_n|}\right) \to 0$$

for all $g \in G$.

Theorem: Let G be a group. Then, G is amenable if and only if G admits a Følner sequence.

Proof. Suppose G admits a Følner sequence. Then, selecting a nonprincipal ultrafilter ω on $\mathbb{N}$, we have that

the map

$$m(f) = \lim_{\omega} \frac{1}{|F_n|} \sum_{h \in F_n} f(h)$$

is a mean, following from the fact that, for all $g \in G$, we have

$$\begin{aligned}
|m(g.f) - m(f)| &= \lim_{\omega} \left| \frac{1}{|F_n|} \left(\sum_{h \in F_n} f\left(g^{-1}h\right) - \sum_{h \in F_n} f(h) \right) \right| \\
&= \lim_{\omega} \left| \frac{1}{|F_n|} \left(\sum_{h \in gF_n} f(h) - \sum_{h \in F_n} f(h) \right) \right| \\
&= \lim_{\omega} \left| \frac{1}{|F_n|} \sum_{h \in gF_n \triangle F_n} f(h) \right| \\
&\leq \lim_{\omega} \|f\|_{\ell^\infty} \frac{|gF_n \triangle F_n|}{|F_n|} \\
&= 0
\end{aligned}$$

since limits along ultrafilters are an extension of standard limits.

In the reverse direction, we note that since each F_n is finite, if we define $f_n = \chi_{F_n}$ to be their respective indicator functions, then $(f_n)_n$ is an approximate mean, which we have shown to be equivalent to a mean by the earlier results. □

This paired with the many previously established results gives us a wide variety of amenable groups.

- The sets

  $$F_n = \{-n, -n+1, \ldots, n-1, n\}$$

  form a Følner sequence in $\mathbb{Z}$. Therefore, $\mathbb{Z}$ is amenable.
- Since any finitely generated abelian group is the product of factors of $\mathbb{Z}$ and finite groups, it follows that any finitely generated abelian group is amenable.
- Since any abelian group is the inductive limit of its finitely generated groups, any abelian group is amenable.
- Similarly, any solvable group can be written as a composition series

  $$G = G_0 \trianglerighteq G_1 \trianglerighteq G_2 \trianglerighteq \cdots \trianglerighteq G_n = \{e\}$$

  with cyclic factors. Since G_{n-1}/G_n is cyclic (hence abelian) and G_n is finite, it follows that G_{n-1} is amenable, and inductively, we get that G is amenable.
- Since any nilpotent group is solvable, it follows that any nilpotent group is amenable.

Proposition: Let G be a finitely generated group with subexponential growth. Then, the sets $(B_S(e,n))_n$ contain a Følner sequence.

Proof. Since the growth rate map $n \overset{\beta}{\mapsto} |B_S(e,n)|$ has subexponential growth, it follows that for all $\varepsilon > 0$, and for all $r, N \in \mathbb{N}$, there exists $n \geq N$ such that

$$\frac{\beta(n+r)}{\beta(n)} < \varepsilon.$$

Inductively, this allows us to find a strictly increasing sequence $\left(n_j\right)_j$ with

$$\frac{\beta\left(n_j+j\right)}{\beta\left(n_j\right)} < 1+\frac{1}{j}.$$

We will show that $\left(F_{n_j}\right)_j$ is a Følner sequence.

For any $g \in B_S(e,r)$, we have that $gB_S(e,n)\triangle B_S(e,n) \subseteq B_S(e,n+r)\setminus B_S(e,n)$. Therefore, we have

$$\begin{aligned}|gB_S(e,n)\triangle B_S(e,n)| &\leq |B_S(e,n+r)\setminus B_S(e,n)|\\ &= \beta(n+r)-\beta(n).\end{aligned}$$

Thus, for all $j \geq r$, we have

$$\begin{aligned}\frac{\left|gB_S\left(e,n_j\right)\triangle B_S\left(e,n_j\right)\right|}{\left|B_S\left(e,n_j\right)\right|} &\leq \frac{\beta\left(n_j+r\right)-\beta\left(n_j\right)}{\beta\left(n_j\right)}\\ &= \frac{\beta\left(n_j+r\right)}{\beta\left(n_j\right)}-1\\ &\leq \frac{\beta\left(n_j+j\right)}{\beta\left(n_j\right)}-1\\ &\leq 1+\frac{1}{j}-1\\ &= \frac{1}{j}.\end{aligned}$$

Since any finite subset $E \subseteq G$ is contained in a ball $B_S(e,r)$ for satisfactorily large r, it follows that $\left(F_{n_j}\right)_j$ is a Følner sequence. □

Corollary: Groups of subexponential growth are amenable.

The converse is not necessarily true. One may construct a semidirect product $\mathbb{Z}^2 \rtimes_A \mathbb{Z}$ where $\mathbb{Z}$ acts on $\mathbb{Z}^2$ via the matrix

$$A = \begin{pmatrix}1 & 1\\ 1 & 2\end{pmatrix}.$$

This group has exponential growth, but it is solvable.

Remark: The Milnor–Wolf Theorem provides a dichotomy for solvable groups; specifically, it gives that a solvable group is either exponential growth or it is virtually nilpotent. Furthermore, Gromov's celebrated theorem says that any group is virtually nilpotent if and only if it is of polynomial growth.

Quasi-Isometry Invariance

The fact that groups of subexponential growth are amenable suggests that amenability is a geometric property. This is actually true, and we will prove this geometrically.

Definition: Let G be a finitely generated group with metric d_S. Then, if $F \subseteq G$ is a subset, then the boundary $\partial(F,r)$ is given by $B_S(F,r)\setminus F$.

Følner sequences are then given by sequences $(F_n)_n$ such that

$$\lim_{n\to\infty}\frac{|\partial(F_n,r)|}{|F_n|} = 0.$$

One has $\partial(F,1) = FS \triangle F$.

Theorem: Let G and H be groups with word metrics d_S and d_T with respect to finite symmetric generating sets S and T. If G and H are quasi-isometric, then G is amenable if and only if H is amenable.

Proof. Our goal is to show that Følner sequences are preserved under quasi-isometries. Let $f : G \to H$ be a quasi-isometry, and let H be amenable. There are constants $c, C \in \mathbb{R}_{>0}$ such that

- the map $f : G \to H$ is a (c,c)-quasi-isometric embedding with c-dense image;
- for all finite subsets $F \subseteq H$, we have

$$\begin{aligned} \left|f^{-1}(B_T(F,c))\right| &\geq \frac{1}{C}|F| \\ \left|f^{-1}(F)\right| &\leq C|F|. \end{aligned}$$

Let $(F_n)_n$ be a Følner sequence for H. Then, $\left(\overline{F}_n\right)_n$, defined by

$$\begin{aligned} \overline{F}_n &= f^{-1}(B_S(F_n,c)) \\ &= \{g \in G : d_T(f(g), F_n) \leq c\}. \end{aligned}$$

It follows that

$$\partial\left(\overline{F}_n, r\right) = \left\{g \in G : g \notin \overline{F}_n, d(g, F_n) \leq r\right\}.$$

Let $g \in \partial\left(\overline{F}_n, r\right)$ and let $a \in \overline{F}_n$ implement the inclusion of g. By the definition of f, we have $d_S(f(x), f(y)) \leq cd_T(x,y) + c$ for all x, y, so that

$$\begin{aligned} d(f(x), F_n) &\leq d(f(x), f(a)) + d(f(a), F_n) \\ &\leq (cr + c) + c \\ &= c(r+2). \end{aligned}$$

In particular, we obtain that $f\left(\partial\left(\overline{F}_n, r\right)\right) \subseteq \partial(F_n, c(r+2))$. Using these estimates yields

$$\begin{aligned} \left|\partial\left(\overline{F}_n, r\right)\right| &\leq \left|f^{-1}\left(f\left(\partial\left(\overline{F}_n, r\right)\right)\right)\right| \\ &\leq \left|f^{-1}(\partial(F_n, c(r+2)))\right| \\ &\leq C|\partial(F_n, c(r+2))|, \end{aligned}$$

and

$$\begin{aligned} \left|\overline{F}_n\right| &= \left|f^{-1}(B(F_n,c))\right| \\ &\geq \frac{1}{C}|F_n|, \end{aligned}$$

so that

$$\begin{aligned} \frac{\left|\partial\left(\overline{F}_n, r\right)\right|}{|F_n|} &\leq C^2 \frac{|\partial(F_n, c(r+2))|}{|F_n|} \\ &\xrightarrow{n\to\infty} 0. \end{aligned}$$

Thus, $\left(\overline{F}_n\right)_n$ is a Følner sequence for G. □

More Analytic Characterizations

Theorem: Let G be a (discrete) group. Then, the following are equivalent:

(i) G is amenable;

(ii) for any actions $G \curvearrowright K$ by affine transformations on a compact convex subset of a locally convex space V, there is a fixed point (the fixed point property);

(iii) for any compact Hausdorff space X on which G acts by homeomorphisms, there exists a G-invariant probability measure.

Proof. Let G be amenable. Then, G admits a Følner sequence $(F_i)_i$. Let $p \in K$. For each such i, let

$$x_i = \frac{1}{|F_i|} \sum_{g \in F_i} g.p.$$

Since G acts by affine homeomorphisms and K is convex, it follows that $x_i \in K$ for each i. Choose a cluster point $x \in K$ of the net $(x_i)_i$; without loss of generality, we will let $x = \lim_i x_i$. Then, for all $g \in G$ and all defining seminorms ρ on V, we have

$$\begin{aligned}
\rho(g.x - x) &= \rho\left(\lim_i \frac{1}{|F_i|} \left(\sum_{h \in F_i} (gh).p - \sum_{h \in F_i} h.p \right) \right) \\
&= \lim_i \rho\left(\frac{1}{|F_i|} \sum_{h \in gF_i \triangle F_i} h.p \right) \\
&\leq \lim_i \frac{1}{|F_i|} \sum_{h \in gF_i \triangle F_i} \rho(h.p) \\
&\leq \lim_i \frac{|gF_i \triangle F_i|}{|F_i|} \left(\sup_{x \in K} \rho(x) \right) \\
&\to 0.
\end{aligned}$$

Therefore, since $\rho(g.x - x) = 0$ for all $g \in G$ and all defining seminorms ρ, it follows that $g.x = x$ for all $g \in G$, so that x is our desired fixed point.

Suppose G satisfies the fixed point property. Then, since the set $\mathrm{Prob}(X)$ is a weak*-compact convex subset of $M(X)$ and the action of G is affine on $\mathrm{Prob}(X)$. Therefore, there is a G-invariant probability measure.

Finally, we may view the left action of G on itself as homeomorphisms of the Stone–Čech compactification βG, and we may identify $\ell^\infty(G) \cong C(\beta G)$, and $M(\beta G) \cong \ell^\infty(G)^*$. Therefore, there is a G-invariant probability measure on βG that corresponds to a G-invariant linear functional $m \in \ell^\infty(G)^*$. □

Next, we discuss the results related to operator algebras.

Definition: We say a function $\varphi \colon \Gamma \to \mathbb{C}$ is *positive definite* if the matrix $\left(\varphi\left(s^{-1}t\right)\right)_{s,t \in F} \in \mathbb{M}_F(\mathbb{C})$ is positive for all finite $F \subseteq \Gamma$.

Given a positive definite function φ, and some finitely supported functions $f, g \in C_c(\Gamma)$, we may define a sesquilinear form on $C_c(\Gamma)$ by

$$\langle f, g \rangle = \sum_{s,t \in \Gamma} \varphi\left(s^{-1}t\right) f(t) \overline{g(s)}.$$

This form is positive semidefinite. Modding out by zero elements and completing gives the Hilbert space $\ell^2_\varphi(\Gamma)$. If $f \in C_c(\Gamma)$, then we will let $\hat{f} \in \ell^2_\varphi(\Gamma)$ denote the natural image. We may then define a unitary representation $\lambda^\varphi \colon \Gamma \to B\left(\ell^2_\varphi(\Gamma)\right)$ by $\lambda^\varphi_s\left(\hat{f}\right) = \widehat{s.f}$, where $s.f(t) = f\left(s^{-1}t\right)$ is the left action of Γ on $C_c(\Gamma)$.

One observes that

$$\begin{aligned}\left\langle \lambda_s^{\varphi}\hat{\delta}_e, \hat{\delta}_e\right\rangle &= \left\langle \hat{\delta}_s, \hat{\delta}_e\right\rangle \\ &= \varphi(s)\end{aligned}$$

for all $s \in \Gamma$, meaning that we recover φ from the vector functional $\left\langle \cdot\hat{\delta}_e, \hat{\delta}_e\right\rangle$.

One sees that if $\varphi\colon C^*(\Gamma) \to \mathbb{C}$ is a positive linear functional, then for each $s_1, \dots, s_n \in \Gamma$, one has

$$\left(\varphi\left(s_i^{-1}s_j\right)\right)_{i,j} = (\mathrm{id}_n \otimes \varphi)\left(\begin{pmatrix} s_1 & s_2 & \cdots & s_n \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}^* \begin{pmatrix} s_1 & s_2 & \cdots & s_n \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}\right),$$

so that $\varphi|_\Gamma$ is a positive definite function. Furthermore, the GNS space for $C^*(\Gamma)$ with respect to φ is precisely $\ell^2_\varphi(\Gamma)$.

There are a number of useful applications of positive definite functions.

Proposition: Let $\Lambda \subseteq \Gamma$ be an inclusion of countable groups. Then, there is a canonical inclusion of full group C^*-algebras $C^*(\Lambda) \subseteq C^*(\Gamma)$.

Proof. By the universal property, there is a canonical $*$-homomorphism $\pi\colon C^*(\Lambda) \to C^*(\Gamma)$. The task is to show that this $*$-homomorphism is injective.

Observe that $C^*(\Lambda)$ has a faithful state. We may view this faithful state as a positive definite function on Λ. Extend φ to Γ by setting $\varphi(s) = 0$ for all $s \notin \Lambda$. We will show that this extension is positive definite on Γ, seeing as the GNS representation of $C^*(\Lambda)$ with respect to φ will be a subrepresentation on π composed with the GNS representation of $C^*(\Gamma)$ with respect to the extension of φ to Γ.

Observe that for any finite $F \subseteq \Gamma$, we have $\left(\varphi\left(s^{-1}t\right)\right)_{s,t\in F}$ is block diagonal with respect to the left coset decomposition $\Lambda\backslash\Gamma$. Since a block diagonal matrix is positive if and only if every block is positive, we may assume that $F \subseteq g\Lambda$ for some fixed $g \in \Gamma$. Positive definiteness follows immediately. □

Analogously, one observes that

$$\ell^2(\Gamma) \cong \bigoplus_{g\in E} \ell^2(\Lambda g)$$

for some family of representatives for right cosets E of Γ/Λ. This gives that the left regular representation of Γ is a multiple of the left regular representation of Λ, so that $C^*_\lambda(\Lambda) \subseteq C^*_\lambda(\Gamma)$.

Definition: Let $\varphi\colon \Gamma \to \mathbb{C}$ be a function. We define $\omega_\varphi\colon \mathbb{C}[\Gamma] \to \mathbb{C}$ and $m_\varphi\colon \mathbb{C}[\Gamma] \to \mathbb{C}[\Gamma]$ by

$$\begin{aligned}\omega_\varphi\left(\sum_{t\in\Gamma} \alpha_t t\right) &= \sum_{t\in\Gamma} \alpha_t \varphi(t) \\ m_\varphi\left(\sum_{t\in\Gamma} \alpha_t t\right) &= \sum_{t\in\Gamma} \alpha_t \varphi(t) t.\end{aligned}$$

Theorem: Let $\varphi\colon \Gamma \to \mathbb{C}$ be a function with $\varphi(e) = 1$. Then, the following are equivalent:

(i) φ is positive definite;

(ii) there is a unitary representation λ_φ of Γ on a Hilbert space H_φ with a unit vector ξ_φ such that

$$\varphi(s) = \left\langle \lambda_\varphi(s)\xi_\varphi, \xi_\varphi\right\rangle;$$

(iii) the functional ω_φ extends to a state on $C^*(\Gamma)$;

(iv) the multiplier m_φ extends to a unital completely positive map on either $C^*(\Gamma)$ or $C^*_\lambda(\Gamma)$, or extends to a normal unital completely positive map on $L(\Gamma)$.

Proof. The fact that (i) implies (ii) is simply the GNS theorem for positive definite functions. Similarly (ii) implies (iii) immediately from bootstrapping.

Next, we show that (iii) implies (iv). Identify ω_φ with a vector state in the universal representation $C^*(\Gamma) \subseteq B(H)$, where H is the direct sum of all GNS representations. By Fell's Absorption Principle, there is a conjugation of $C^*_\lambda(\Gamma) \otimes 1$ onto the diagonal consisting of elements of the form

$$\theta = \sum_{t\in\Gamma} \alpha_t(\lambda_t \otimes \lambda_t).$$

In particular, this means that the map

$$\sum_{t\in\Gamma} \alpha_t \lambda_t \overset{\sigma}{\mapsto} \sum_{t\in\Gamma} \alpha_t(\lambda_t \otimes \lambda_t)$$

is a $*$-homomorphism that extends to a normal $*$-homomorphism $L(\Gamma) \to L(\Gamma) \otimes B(H)$. □

Theorem: The following are equivalent:

(i) Γ is amenable;

(ii) there is a net $(\xi_i)_i$ of unit vectors such that $\|\lambda_s \xi_i - \xi_i\|_{\ell^2(\Gamma)} \to 0$;

(iii) there is a net $(\varphi_i)_i$ of finitely supported positive definite functions on Γ such that $\varphi_i \to 1$ pointwise;

(iv) $C^*(\Gamma) = C^*_\lambda(\Gamma)$;

(v) C^*_λ has a character.

Proof. Let $(F_i)_i$ be a Følner sequence, and let $\xi_i = \frac{1}{\sqrt{|F_i|}}\chi_{F_i}$ be normalized characteristic functions of the F_i. Then, we have that

$$\left\| \frac{1}{\sqrt{|F_i|}}\chi_{sF_i} - \frac{1}{\sqrt{|F_i|}}\chi_{F_i} \right\|_{\ell^2(\Gamma)} = \sqrt{\frac{|sF_i \triangle F_i|}{|F_i|}} \to 0.$$

This yields our desired almost-invariant vectors — i.e., (i) implies (ii).

Now, if $(\xi_i)_i$ is a sequence of almost-invariant vectors, then we observe that the vector states $x \overset{\varphi_i}{\longmapsto} \langle x\xi_i, \xi_i\rangle$ restrict to positive definite functions on Γ that tend to 1 pointwise. Since we may approximate the ξ_i by finitely supported functions, we have that (ii) implies (iii).

Taking $(\varphi_i)_i$ is this net of functions, then the multipliers m_{φ_i} on $C^*(\Gamma)$ and $\widetilde{m}_{\varphi_i}$ on $C^*_\lambda(\Gamma)$ are unital and completely positive maps. Since on $\mathbb{C}[\Gamma]$, we have

$$\begin{aligned}
\lambda \circ m_{\varphi_i}\left(\sum_{t\in\Gamma} \alpha_t t\right) &= \lambda\left(\sum_{t\in\Gamma} \alpha_t \varphi_i(t) t\right) \\
&= \sum_{t\in\Gamma} \alpha_t \varphi_i(t) \lambda_t \\
&= \widetilde{m}_{\varphi_i}\left(\sum_{t\in\Gamma} \alpha_t \lambda_t\right) \\
&= \widetilde{m}_{\varphi_i} \circ \lambda\left(\sum_{t\in\Gamma} \alpha_t t\right),
\end{aligned}$$

we have that $\lambda \circ m_{\varphi_i} = \widetilde{m}_{\varphi_i} \circ \lambda$ on $C^*(\Gamma)$ since they're continuous and coincide on the dense subspace $\mathbb{C}[\Gamma]$. Notice that $m_{\varphi_i}(x) \to x$ for every $x \in C^*(\Gamma)$ since it holds for $\mathbb{C}[\Gamma]$. Suppose $x \in C^*(\Gamma)$ and $\lambda(x) = 0$. Then,

$$\begin{aligned}\lambda\Big(m_{\varphi_i}(x)\Big) &= m_{\varphi_i}(\lambda(x)) \\ &= 0\end{aligned}$$

for all i. Yet, since φ_i is finitely supported, we have $m_{\varphi_i}(x) \in \mathbb{C}[\Gamma]$, so $\lambda\Big(m_{\varphi_i}(x)\Big) = 0$ implise that $m_{\varphi_i}(x) = 0$, so $x = \lim_i m_{\varphi_i}(x) = 0$, so $\lambda\colon C^*(\Gamma) \to C^*_\lambda(\Gamma)$ is injective. Thus, (iii) implies (iv).

Next, we observe that $C^*(\Gamma)$ admits a character by using the universal property to extend from the trivial representation. Thus, since $C^*(\Gamma) = C^*_\lambda(\Gamma)$, it follows that $C^*_\lambda(\Gamma)$ admits a character.

Finally, suppose $\tau\colon C^*_\lambda(\Gamma) \to \mathbb{C}$ is a character. We view τ as a state. By the Arveson Extension Theorem, we may extend τ to all of $B\Big(\ell^2(\Gamma)\Big)$, so τ is defined on $\ell^\infty(\Gamma) \subseteq B\Big(\ell^2(\Gamma)\Big)$ viewed as multiplication operators. Notice that, if we let $s.f(t) = f\Big(s^{-1}t\Big)$, we have that

$$\begin{aligned}\lambda_s M_f \lambda_s^*(\delta_t) &= \lambda_s M_f(\delta_{s^{-1}t}) \\ &= \lambda_s(f\delta_{s^{-1}t}) \\ &= (s.f)\delta_t \\ &= M_{s.f}\delta_t.\end{aligned}$$

Therefore, if we let f represent the multiplication operator M_f, we have

$$\begin{aligned}\tau(s.f) &= \tau(\lambda_s f \lambda_s^*) \\ &= \tau(\lambda_s)\tau(f)\tau(\lambda_s^*) \\ &= \tau(f),\end{aligned}$$

where we note that λ_s are in the multiplicative domain of τ. Thus, τ is an invariant mean. □

Soficity

We will work to understand soficity (a type of generalized approximation property for groups) with a variety of perspectives. This will allow us to establish soficity for a variety of circumstances.

The Law of the Group von Neumann Algebra

This is the approach that my advisor, Ben Hayes, takes towards tracial von Neumann algebras, specialized to the case of groups.

A tracial von Neumann algebra is a pair (M, τ), where M is a von Neumann algebra and $\tau\colon M \to \mathbb{C}$ is a linear functional that satisfies:

- positivity: $\tau(x^*x) \geq 0$ for all $x \in M$;
- normalization: $\tau(1) = 1$;
- traciality: for all $x, y \in M$, $\tau(xy) = \tau(yx)$.

If I is any index set, an I-tuple $a \in M^I$ yields a $*$-algebra homomorphism from the space of all noncommutative $*$-polynomials, denoted $\mathbb{C}^*\langle (T_i)_{i\in I}\rangle$, given by $P(a) \in M$ for all $P \in \mathbb{C}^*\langle (T_i)_{i\in I}\rangle$.

The *law* of an I-tuple $a \in M^I$ is then a linear functional $\ell_a\colon \mathbb{C}^*\langle (T_i)_{i\in I}\rangle \to \mathbb{C}$ given by

$$\ell_a(P) = \tau(P(a)).$$

Notice that in the case of a single-element set for I and a normal or self-adjoint element for A, the law is given by integration with respect to the spectral measure of a,

$$\ell_a(P) = \int P \, d\mu_a.$$

In a sense, the law is a generalization of the spectral measure.

Now, consider a group G with finite symmetric generating set S. Then, in the group von Neumann algebra $L(G)$, there is a distinguished tuple $(\lambda_s)_{s\in S} \in L(G)^S$ given by the left-translation operators λ_s for each $s \in S$. The corresponding law is then given by

$$\ell_S(P) = \langle P((\lambda_s)_{s\in S})\delta_e, \delta_e\rangle.$$

We start by viewing the symmetric group $\mathrm{Sym}(n) \subseteq U(\mathbb{M}_n(\mathbb{C}))$ as permutation matrices. Then, any S-tuple $\sigma \in \mathrm{Sym}(n)^S$ has law given by

$$\ell_\sigma(P) = \mathrm{tr}_n(P((\sigma_s)_{s\in S})).$$

Notice that, given a word w in $\mathbb{F}_S$, which we can view as a monomial in $\mathbb{C}^*\langle (T_i)_{i\in I}\rangle$, we observe that the non-normalized trace is the cardinality of the fixed-point set, so that

$$\mathrm{tr}_n(w((\sigma_s)_{s\in S})) = \frac{1}{n}\Big|\{j \in [n] : w((\sigma_s)_{s\in S})(j) = j\}\Big|.$$

We then say that G is sofic if and only if there is a sequence of natural numbers $(k_n)_n$ and a corresponding sequence of S-tuples $(\sigma_n)_n$ with each $\sigma_n \in \mathrm{Sym}(k_n)^S$, such that the laws ℓ_{σ_n} converge weak* (i.e., pointwise) to the law ℓ_S.

Metric Ultraproducts

The original notion of an ultraproduct came from consideration of normed spaces and a development of nonstandard analysis.

Let E_α be a family of normed spaces indexed by a set A. We may form the ℓ^∞-direct product of the set $\{E_\alpha : \alpha \in A\}$ by taking

$$\prod_{\alpha\in A} E_\alpha := \left\{(x_\alpha)_{\alpha\in A} : x_\alpha \in E_\alpha \text{ and } \sup_\alpha \|x_\alpha\| < \infty\right\},$$

and we may equip it with a norm

$$\|x\| = \sup_{\alpha\in A}\|x_\alpha\|_\alpha.$$

Write $\mathcal{E}$ for the ℓ^∞-direct product.

Given a (non-principal) ultrafilter $\mathcal{U}$ on A, there is a way to define a "null subspace" by taking

$$\mathcal{N}_\mathcal{U} = \left\{x \in \mathcal{E} : \lim_\mathcal{U}\|x_\alpha\| = 0\right\},$$

where the limit along the ultrafilter $\mathcal{U}$ of a sequence $(y_\alpha)_\alpha$ is defined to be y if for all $\varepsilon > 0$, the set $\left\{\alpha : \left\|y_\alpha - y\right\| < \varepsilon\right\}$ is contained in $\mathcal{U}$. The subspace $\mathcal{N}_\mathcal{U}$ is closed. The metric ultraproduct of the family $(E_\alpha)_{\alpha\in A}$ is then given by

$$\prod_\mathcal{U} E_\alpha := \mathcal{E}/\mathcal{N}_\mathcal{U},$$

with norm given by

$$\left\|[x]_{\mathcal{U}}\right\| = \lim_{\mathcal{U}} \|x_\alpha\|.$$

We write E for the ultraproduct. When the ultrafilter is nonprincipal, then the ultraproduct is complete. If $(x_k)_k$ is a Cauchy sequence of elements of E, then for each $i \in \mathbb{N}$, we may find some N_i such that

$$\left\|x'_N - x_N\right\| < 2^{-i}$$

for all $N, N' \geq N_i$. For each k, we select a representative $(x_k(\alpha))_{\alpha \in A}$, and define

$$I_i = \left\{\alpha \in A : \left\|x_{N_i}(\alpha) - x_{N_{i+1}}(\alpha)\right\| < 2^{-i}\right\}.$$

We have that each I_i is contained in $\mathcal{U}$, and without loss of generality we may assume that $I_1 \supseteq I_2 \supseteq \cdots$ with empty intersection, since the ultrafilter $\mathcal{U}$ is non-principal, and thus is countably incomplete. Define an element $x \in \mathcal{E}$ by taking

$$x|_{I_i \setminus I_{i+1}} = x_{N_i}|_{I_i \setminus I_{i+1}}.$$

The equivalence class $[x]_{\mathcal{U}}$ is the limit of the Cauchy sequence $(x_k)_k$.

Next, we concern ourselves with ultraproducts of metric groups. Given a family (G_α, d_α) of topological groups with left-invariant metrics,[II] we may form an ultraproduct by following an analogous process for the case of metric spaces, but the result is only a metric space on which the Cartesian product group $\prod_{\alpha \in A} G_\alpha$ acts by isometries (i.e., a homogeneous space), rather than a proper group.

> **Example:** Consider the symmetric group $\mathrm{Sym}(\mathbb{N})$ on the natural numbers. The standard Polish topology on $\mathrm{Sym}(\mathbb{N})$ is the subspace topology induced by the product topology on the prodiscrete space $\mathbb{N}^{\mathbb{N}}$. This is a separable group topology with a compatible metric given by
>
> $$d(\sigma, \tau) = \sum_{i=1}^{\infty} \left\{2^{-i} : \sigma(i) \neq \tau(i)\right\}.$$
>
> We note that this is a bounded analogue of the Hamming distance for finite symmetric groups.
>
> We will form an ultrapower of the group $(\mathrm{Sym}(\mathbb{N}), d)$ with regard to a nonprincipal ultrafilter $\mathcal{U}$ on $\mathbb{N}$. Since the metric d is itself bounded, we see that the space $\mathcal{G} = (\mathrm{Sym}(\mathbb{N}))^{\mathbb{N}}$ is the analogue of the space $\mathcal{E}$ from the construction of the ultraproduct of normed spaces.
>
> The set
>
> $$\mathcal{N} = \left\{x : \lim_{\mathcal{U}} d(x_\alpha, e) = 0\right\}$$
>
> is a subgroup of $\mathcal{G}$, seeing as the invariance of the metric d yields
>
> $$\begin{aligned} d(xy, e) &= d\left(y, x^{-1}\right) \\ &\leq d(y, e) + d\left(x^{-1}, e\right) \\ &= d(y, e) + d(x, e). \end{aligned}$$
>
> Yet, it is not normal. If we have transpositions $x = (x_i)_i = ((i, i+1))_{i \in \mathbb{N}}$ and $y = (y_i)_i = ((1, i))_{i \in \mathbb{N}}$, then $x \in \mathcal{N}$ but $y^{-1}xy \notin \mathcal{N}$, since $y_i^{-1} x_i y_i = (1, i+1)$ for each i, and we have $d\left(y_i^{-1} x_i y_i, e\right) \to 1/2$.
>
> In particular, this means that the homogeneous space $\mathcal{G}/\mathcal{N}$ admits a $\mathcal{G}$-invariant metric $d(x, y) = \lim_{\mathcal{U}} d(x_n, y_n)$, it is not a group.

[II] Such invariant metrics always exist for metrizable groups by a theorem of Kakutani.

In order to develop an ultraproduct of metric groups that is actually a group, we need to use bi-invariant metrics. That is, metrics that satisfy $d(x,y) = d(gx,gy) = d(xg,yg)$. These metrics are special because they admit topologies whose conjugation-invariant open subsets form a neighborhood basis at the identity.

If $(G_\alpha, d_\alpha)_\alpha$ are a family of metric groups with bi-invariant metrics, then the "null set" subgroup $\mathcal{N}$ is normal, and for any ultrafilter $\mathcal{U}$ on A, the ultraproduct

$$\prod_{\mathcal{U}} G_\alpha = \mathcal{G}/\mathcal{N}$$

is indeed a group. The bi-invariant metric

$$d(x\mathcal{N}, y\mathcal{N}) = \lim_{\mathcal{U}} d_\alpha(x_\alpha, y_\alpha)$$

yields the metric ultraproduct of the family (G_α, d_α) with respect to the ultrafilter $\mathcal{U}$.

Example (Groups with Bi-Invariant Metrics)**:**

(i) If (X,μ) is a finite measure space, then group $\mathrm{Aut}(X,\mu)$ admits a bi-invariant metric given by

$$d(\sigma,\tau) = \mu(\{x \in X : \sigma(x) \neq \tau(x)\}).$$

(ii) The (normalized) Hamming distance on the symmetric group S_n is given by

$$d_H(\sigma,\tau) = \frac{1}{n}|\{i \in [n] : \sigma(i) \neq \tau(i)\}|.$$

This is the special case of a finite set $[n]$ equipped with the (normalized) counting measure.

(iii) If H is a Hilbert space, then $U(H)$ denotes the group of unitary operators on H. The *uniform operator metric* is given by

$$d_u(u,v) = \|u - v\|.$$

This is a bi-invariant metric.

(iv) If $H = \mathbb{C}^n$ is an n-dimensional Hilbert space, then $U(H) = U(n)$ is the unitary group of rank n. The *normalized Hilbert–Schmidt metric* on $U(n)$ is given by

$$d_{\mathrm{HS}}(u,v) = \sqrt{\mathrm{tr}_n((u-v)^*(u-v))},$$

where tr_n is the normalized trace $\mathrm{tr}_n = \frac{1}{n}\mathrm{Tr}$.

Every metric group G embeds into its ultrapower $G^{\mathcal{U}}$ with the diagonal embedding.

Definition: A group G is called sofic if it admits an embedding into a metric ultraproduct of certain symmetric groups. That is, there is some set A, finite subsets $(F_\alpha)_{\alpha\in A}$ indexed by A, and some nonprincipal ultrafilter $\mathcal{U}$ on A such that

$$G \hookrightarrow \prod_{\mathcal{U}} \left(\mathrm{Sym}(F_\alpha), d_{F_\alpha}\right)$$

is a well-defined embedding.

Approximate Homomorphisms

The metric ultraproduct definition is equivalent to another definition that shows soficity is a special type of approximation property. This will enable us to establish the soficity of a large class of groups.

Definition: Let G be a group, $K \subseteq G$ a finite subset, and $\varepsilon > 0$. A map $\varphi\colon G \to \mathrm{Sym}(F)$ for some finite set F is called a (K,ε)-almost-homomorphism if it satisfies the following conditions:

(i) for all $k_1, k_2 \in K$ with $k_1 k_2 \in K$, we have $d_F(\varphi(k_1 k_2), \varphi(k_1)\varphi(k_2)) \leq \varepsilon$;

(ii) for all $k_1, k_2 \in K$ with $k_1 \neq k_2$, we have $d_F(\varphi(k_1), \varphi(k_2)) \geq 1 - \varepsilon$.

Here, d_F denotes the normalized Hamming distance on $\mathrm{Sym}(F)$.

Definition: A group G is called sofic if, for every finite subset $K \subseteq G$ and every $\varepsilon > 0$, there is a nonempty finite set F and a (K, ε)-almost-homomorphism $\varphi\colon G \to \mathrm{Sym}(F)$.

We will now discuss the equivalence of this definition and the metric ultraproduct definition.

Theorem: The metric ultraproduct definition of a sofic group and the almost-homomorphism definition of a sofic group coincide, where we note that we only need to assume that for all $k_1, k_2 \in K$ with $k_1 \neq k_2$, we have $d_F(\varphi(k_1), \varphi(k_2)) \geq 1/4$ in the second item.

Proof. We will always assume that G is countable. In one direction, we let $\mathcal{K}$ be the set of finite subsets of G ordered by inclusion.

Then, we will consider the family of $(K, 1/n)$-homomorphisms $\varphi_{K,1/n}\colon (K, 1/n) \to \mathrm{Sym}(F_{K,1/n})$ for every finite set $K \subseteq G$ and strictly positive natural number n. Choose a nonprincipal ultrafilter $\mathcal{U}$ on the set of all such pairs such that $\mathcal{U}$ contains all pairs of the form $(K', 1/m)$ with $K' \supseteq K$ and $m \geq n$. The map

$$\varphi\colon G \to \prod_{\mathcal{U}} \left(\mathrm{Sym}(F_{K,1/n}), d_{F_{K,1/n}}\right)$$
$$g \mapsto (\varphi_{K,1/n}(g))_{\mathcal{U}}$$

is a well-defined group homomorphism satisfying $d(\varphi(g), \varphi(h)) \geq 1/4$ for any $g \neq h$.

In the reverse direction, the given embedding $\varphi\colon G \to \prod_{\mathcal{U}} \mathrm{Sym}(F_\alpha)$ yields a family of (K, ε)-almost-homomorphisms by choosing satisfactory coordinate projections for the given choices of K and ε. For the final condition, we note that choices of coordinate projections may not necessarily yield $d(\varphi(g), \varphi(h)) \geq 1/4$ whenever $g, h \in K$ with $g \neq h$. To resolve this issue, we amplify as follows.

Fix a (K, ε')-almost-homomorphism $\theta\colon F \to S_n$ with $\theta(g) \neq \theta(h)$ and ε' sufficiently small. Let $\delta = d(\theta(g), \theta(h))$. We amplify by embedding K diagonally into the group of permutations of the square $[n] \times [n]$ diagonally; that is,

$$\theta^{(2)}(g) = (\theta(g), \theta(g))$$

acting by double permutations. Then, the normalized counting measure of the set of pairs (i, j) for which $(\theta(g), \theta(g))$ and $(\theta(h), \theta(h))$ are different is strictly larger than δ. Applying this amplification method as many times as necessary yields our desired separation of $1/4$.

Since we may choose δ to be arbitrarily close to $d(g, h)$ as we desire, the amplification level k can be chosen before we choose the (K, ε')-almost-homomorphism by letting ε' be small enough such that $\theta^{(k)}$ remains a ε-almost-homomorphism for prescribed $\varepsilon > 0$. □

We begin establishing the soficity of large classes of groups.

Theorem: Locally sofic groups — that is, groups whose finitely generated subgroups are sofic — is sofic.

Proof. Let $K \subseteq G$ be a finite subset, and let $\varepsilon > 0$. Let $H = \langle K \rangle$ be the finitely generated subgroup of G generated by K. Then, since H is sofic, there is a (K, ε)-almost-homomorphism $\psi\colon H \to \mathrm{Sym}(F)$ for some nonempty finite set F. We may extend ψ arbitrarily to a map $\varphi\colon G \to \mathrm{Sym}(F)$ setting $\varphi(g) = \mathrm{id}_F$ for all $g \in G \setminus H$, and note that φ remains a (K, ε)-almost-homomorphism. □

In particular, every locally finite group is sofic. This is also established by the following more general proposition.

Proposition: Amenable groups are sofic.

Proof. Let G be amenable, $K \subseteq G$ finite, and $\varepsilon > 0$. Let $S = \left(\{e_G\} \cup K \cup K^{-1}\right)^2$. Since G is amenable, the

Følner criterion gives a nonempty finite subset $F \subseteq G$ such that

$$|F \setminus sF| \leq \frac{\varepsilon}{|S|}|F|$$

for all $s \in S$. Let

$$E = \bigcap_{s \in S} sF.$$

Note that $E \subseteq F$ since $e_G \in S$. Since $S = S^{-1}$, it follows that $sE \subseteq F$ for all $s \in S$, and we have

$$\begin{aligned} |F \setminus E| &= \left| \bigcup_{s \in F} (F \setminus sF) \right| \\ &\leq \sum_{s \in S} |F \setminus sF| \\ &\leq \varepsilon |F|. \end{aligned}$$

Therefore, we have $|E| \geq (1 - \varepsilon)|F|$.

For every $g \in G$, we have $|F| = |gF|$, so $|F \setminus gF| = |gF \setminus F|$, and we may find a bijective map $\alpha_g \colon gF \setminus F \to F \setminus gF$. Let $\varphi \colon G \to \mathrm{Sym}(F)$ be defined by

$$\varphi(g)(f) = \begin{cases} gf & gf \in F \\ \alpha_g(gf) & \text{else} \end{cases}$$

for any $g \in G$ and $f \in F$.

If $k_1, k_2 \in K$ and $f \in E$, then we have $k_2, k_1 k_2 \in S$, meaning $k_2 f, k_1 k_2 f \in F$, meaning $\varphi(k_1 k_2)(f) = k_1 k_2 f$, and

$$\begin{aligned} (\varphi(k_1)\varphi(k_2))(f) &= \varphi(k_1)(\varphi(k_2)(f)) \\ &= \varphi(k_1)(k_2 f) = k_1 k_2 f. \end{aligned}$$

Therefore, $\varphi(k_1 k_2)$ and $\varphi(k_!)\varphi(k_2)$ coincide on E, so by the definition of E we have

$$\begin{aligned} d_F(\varphi(k_1 k_2), \varphi(k_1)\varphi(k_2)) &\leq \frac{|F \setminus E|}{|F|} \\ &\leq \varepsilon. \end{aligned}$$

Meanwhile, if $k_1, k_2 \in K$ with $k_1 \neq k_2$, and $f \in E$, then we have $k_1 f, k_2 f \in F$, so that $\varphi(k_1)(f) = k_1 f \neq k_2 f = \varphi(k_2) f$. Thus, we have

$$\begin{aligned} d_F(\varphi(k_1), \varphi(k_2)) &\geq \frac{|E|}{|F|} \\ &\geq 1 - \varepsilon. \end{aligned}$$

This gives that G is a sofic group. □

Cayley Graphs

Given a group G, we have established previously that we only need to verify that all its finitely generated subgroups are sofic in order to establish that G is sofic.

This will allow us to consider a classification of soficity of a finitely generated subgroup by its relationship to the Cayley graph.

Let G be a finitely generated group with finite symmetric generating set S. For any $r > 0$, denote by $B_S(r)$ the ball of radius r centered at the identity element. We may label the edges of the graph by S — i.e., labeling by s the edge $\{g, gs\}$. Let $X = (V, E)$ be an S-labeled graph. Given $q \in V$ and $r \in R$, let $B(q, r)$ be the ball of radius r centered at the vertex q with the induced S-labeled graph structure.

Let $Q(r)$ be the set of all $q \in V$ such that there exists an isomorphism of S-labeled graphs $\psi_{q,r} \colon B_S(r) \to B(q, r)$ satisfying $\psi_{q,r}(e_G) = q$.

If such a map $\psi_{q,r}$ exists, then it is unique. There is a sequence of inclusions

$$V \supseteq Q(0) \supseteq Q(1) \supseteq \cdots \supseteq Q(r) \supseteq Q(r+1) \supseteq \cdots,$$

where we note that $\mathrm{Cay}(G, S)$ is edge-symmetric with respect to $s \mapsto s^{-1}$, meaning that we will require $B(q, r)$ to be edge-symmetric as well (since $\psi_{q,r}$ is an isomorphism of S-labeled graphs).

Lemma: Let $X = (V, E)$ be an S-labeled graph, $r_0, i \in \mathbb{N}$. Suppose $q_0 \in Q((i+1)r_0)$. Then, $B(q_0, r_0) \subseteq Q(ir_0)$.

Proof. Let $q' \in B(q_0, r_0)$. We will show that $q' \in Q(ir_0)$. From the triangle inequality, we have that $B(q', ir_0) \subsetneq B(q_0, (i+1)r_0)$. Since $\psi_{q_0,(i+1)r_0}$ is isometric, we have that $g = \psi^{-1}_{q_0,(i+1)r_0}(q')$ is contained in $B_S(e, r_0)$. In particular, this means that $gh \in B((i+1)r_0)$. Therefore, the map ψ_{q',ir_0} defined by taking $\psi_{q',ir_0}(h) = \psi_{q_0,(i+1)r_0}(gh)$ for all $h \in B_S(e, ir_0)$ yields an S-labeled graph isomorphism with $\psi_{q',ir_0}(e) = q'$. Therefore, $q' \in Q(ir_0)$, so that $B(q_0, r_0) \subseteq Q(ir_0)$. □

Lemma: Let $G = (V, E)$ be an S-labeled graph, and let $r_0 \in \mathbb{N}$. Let $q_1, q_2 \in Q(2r_0)$ be such that $q_1 \neq q_2$ and $g \in B_S(e, r_0)$. Then,

$$\psi_{q_1,2r_0}(g) \neq \psi_{q_2,2r_0}(g).$$

Proof. If $g = e_G$, then since $q_1 \neq q_2$, we obtain the desired result.

Suppose $g \neq e_G$, and assume toward contradiction that $\psi_{q_1,2r_0}(g) = \psi_{q_2,2r_0}(g)$. Call this value q_0. Since $g \in B_S(e, r_0)$, there is $1 \leq r' \leq r_0$ and elements $s_1, \ldots, s_{r'}$ in S such that $g = s_1 s_2 \cdots s_{r'}$. Let

$$\pi = [e_G, s_1, s_1 s_2, \ldots, s_1 \cdots s_{r'-1}, g]$$

be a path traversing from e_G to g. Notice that π is contained in $B_S(e, r_0)$. The path π is mapped by $\psi_{q_1,2r_0}$ and $\psi_{q_2,2r_0}$ to paths π_1 and π_2 with initial vertices $\pi_1^- = q_1$ and $\pi_2^- = q_2$ with common terminal vertex $\pi_1^+ = \pi_2^+ = q_0$. Since $\psi_{q_1,2r_0}$ and $\psi_{q_2,2r_0}$ are isometric, π_1 and π_2 are contained in $B(q_0, r_0)$.

Moreover, since $\psi_{q_1,2r_0}$ and $\psi_{q_2,2r_0}$ are label-preserving, it follows that π_1 and π_2 have the same labels; namely, their edges are labeled by $s_1 \cdots s_{r'}$.

The inverse images of π_1 and π_2 under the S-labeled graph isomorphism $\psi_{q_0,r_0} \colon B_S(e, r_0) \to B(q_0, r_0)$ are equal to π. Yet, this means that $\pi_1 = \pi_2$, so $\pi_1^- = \pi_2^-$, or that $q_1 = q_2$, which is a contradiction. Thus, $\psi_{q_1,2r_0}(g) \neq \psi_{q_2,2r_0}(g)$. □

Lemma: Let $G = (V, E)$ be an S-labeled graph, and let $r_0 \in \mathbb{N}$. Let $h, k \in B_S(e, r_0)$, and let $q_0 \in Q(2r_0)$. Then, $\psi_{q_0,2r_0}(h) \in Q(r_0)$.

Writing $\psi_{q_0,2r_0}(h) = v_0$, we have

$$\psi_{q_0,2r_0}(hk) = \psi_{v_0,r_0}(k).$$

Proof. The map $\psi_{q_0,2r_0}$ is isometric, so $\psi_{q_0,2r_0}(h) \in B(q_0, r_0)$. Therefore, since $q_0 \in Q(2r_0)$, it follows that $B(q_0, r_0) \subseteq Q(r_0)$, meaning that $\psi_{q_0,2r_0}(h) \in Q(r_0)$.

Since $hk \in B_S(e, 2r_0)$, it follows that the notion of $\psi_{v_0,r_0}(k)$ is well-defined. If $k = e_G$, then

$$\begin{aligned} \psi_{q_0,2r_0}(hk) &= \psi_{q_0,2r_0}(h) \\ &= \psi_{v_0,r_0}(e_G) \\ &= \psi_{v_0,r_0}(k). \end{aligned}$$

Suppose $k \neq e_G$. Find an S-labeled path π_1 with length $r' \leq r_0$ from $\pi_1^- = h$ to $\pi_1^+ = hk$ contained in $B_S(h, r_0)$, where we recall that π_1^- denotes the start point of the path and π_1^+ denotes the end point of the path. This is contained in $B_S(2r_0)$ since $h, k \in B_S(r_0)$. The image $\overline{\pi_1} = \psi_{q_0,2r_0}(\pi_1)$ is an S-labeled path with length equal to the length of π_1, meaning that $\overline{\pi_1} \subseteq B(q_0, r_0)$.

We have that $v_0 \in Q(r_0)$. The inverse image of the path $\overline{\pi_1}$ under the isomorphism ψ_{v_0,r_0} preserves labels and maps $v_0 = \psi_{q_0,2r_0}(h)$ to e_G. Therefore, we have that the unique S-labeled path π_2 from $\pi_2^- = e_G$ to $\pi_2^+ = k$ is the inverse image $\psi_{v_0,r_0}^{-1}(\overline{\pi_1})$.

In particular, we have that

$$\begin{aligned}\psi_{v_0,r_0}^{-1}\left(\psi_{q_0,2r_0}(hk)\right) &= \psi_{v_0,r_0}^{-1}\left(\overline{\pi_1}^+\right)\\ &= \pi_2^+\\ &= k.\end{aligned}$$

□

This will finally allow us to prove the following theorem.

Theorem: Let G be a finitely generated group, and let S be a finite symmetric generating set for G. The following are equivalent:

(i) G is sofic;

(ii) for all $\varepsilon > 0$ and for all $r \in \mathbb{N}$, there exists a finite S-labeled graph $X = (V, E)$ such that

$$|Q(r)| \geq (1 - \varepsilon)|V|,$$

where $Q(r)$ is the set of all $q \in V$ such that there exists an isomorphism of S-labeled graphs $\psi_{q,r} \colon B_S(e, r) \to B(q, r)$ taking $e \mapsto q$.

Proof. Suppose G is sofic. Let $\varepsilon > 0$ and $r \in \mathbb{N}$ be fixed. Let $K = B_S(2r + 1)$, and let

$$\varepsilon' = \frac{\varepsilon}{1 + |S||B_S(r)| + |B_S(r)|^2}.$$

Since G is sofic, there is a nonempty finite set F and a (K, ε')-almost-homomorphism $\varphi \colon G \to \mathrm{Sym}(F)$. We define an S-labeled graph $X = (V, E)$ by taking $V = F$ and E to be edges $\left\{q, \varphi\left(s^{-1}\right)(q)\right\}$ for each $s \in S$ and each $q \in F$; such an edge will be labeled by s. That is, edges labeled by s are those that connect vertices q to vertices $\varphi\left(s^{-1}\right)(q)$.

For each $q \in V$, let $\psi_q \colon G \to V$ be the map defined by taking $\psi_q(g) = \varphi\left(g^{-1}\right)(q)$ for each $g \in G$. Let $V_0 \subseteq V$ consists of all $q \in V$ satisfying

- $\psi_q(e_G) = q$;
- $\psi_q(gs) = \psi_{\psi_q(g)}(s)$ for all $g \in B_S(r)$ and all $s \in S$;
- $\psi_q(g) \neq \psi_q(h)$ for all $g \neq h$ in $B_S(r)$.

Now, let $q \in V_0$, and let $g \in B_S(r)$. Writing $e_G \neq g \in B_S(r)$ as $g = s_1 \cdots s_{r'}$ where $r' \leq r$, we may consider edges $e_1, \ldots, e_{r'}$ defined by

$$\begin{aligned}e_j &= \left\{\psi_q\left(s_1 \cdots s_{j-1}\right), \varphi\left(s_j^{-1}\right)\left(\psi_q\left(s_1 \cdots s_{j-1}\right)\right)\right\}\\ &= \left\{\psi_q\left(s_1 \cdots s_{j-1}\right), \psi_q\left(s_1 \cdots s_j\right)\right\},\end{aligned}$$

labeled by s_j. The path $\pi = [e_1, \ldots, e_{r'}]$ connects q to $\psi_q(g)$, and has length at most r, giving that $\psi_q(B_S(e, r)) \subseteq B(q, r)$. For the reverse inclusion, if $q' \in B(q, r)$ is equal to q, then $q' = \psi_q(e_G) \in \psi_q(B_S(e, r))$, while if $q' \neq q$, then there is a sequence of edges $[f_1, \ldots, f_{r'}]$ labeled by $s_1, \ldots, s_{r'}$ respectively yielding $q' = \psi_1(s_1 \cdots s_r')$. Thus, $B(q, r) = \psi_q(B_S(e, r))$.

Moreover, the map ψ_q restricted to $B_S(r)$ is injective by definition, so we have that if $g \in B_S(r)$ and $s \in S$, then

$$\begin{aligned}\left\{\psi_q(g), \psi_q(gs)\right\} &= \left\{\psi_q(g), \psi_{\psi_q(g)}(s)\right\} \\ &= \left\{\psi_q(g), \varphi\left(s^{-1}\right)\left(\psi_q(g)\right)\right\} \\ &\in E.\end{aligned}$$

Therefore, the map $\psi_{q,r}\colon B_S(e,r) \to B(q,r)$ is injective. Thus, $V_0 \subseteq Q(r)$.

Our next goal is to estimate the cardinality of V_0. Let d_V be the normalized Hamming distance on $\mathrm{Sym}(V)$.

We start by observing that $d_V(\mathrm{id}_V, \varphi(e_G)) \leq \varepsilon'$ by the definition of a (K,ε')-almost-homomorphism. Furthermore, by the definition of an almost-homomorphism, we have that

$$\begin{aligned}d_V(\varphi(e_G), \varphi(e_G)\varphi(e_G)) &= d_V(\varphi(e_G e_G), \varphi(e_G)\varphi(e_G)) \\ &\leq \varepsilon'\end{aligned}$$

□

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